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Graduation Project

Smart Face Recognition Attendance System

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Abstract

The **Smart Face Recognition Attendance System** is an intelligent solution designed to automate and improve the attendance management process in educational institutions. Traditional attendance methods, such as manual signatures, verbal roll calls, and paper-based records, are time-consuming, prone to human error, and vulnerable to proxy attendance, reducing the accuracy and reliability of attendance records.

The proposed system addresses these challenges by integrating **Face Recognition** and **QR Code** technologies into a unified mobile application. Students can create accounts, register their facial data, enroll in available courses, and generate personal QR codes. During attendance sessions, academic staff can verify attendance quickly and securely using either Face Recognition or QR Code scanning, while the system automatically records attendance and prevents duplicate or fraudulent entries.

The application also enables students to view their enrolled courses, attendance history, and attendance percentages. Academic staff can create courses, manage attendance sessions, monitor attendance records, and generate detailed attendance reports. The backend communicates through secure **RESTful APIs**, while all data is stored in a centralized database with role-based access control to ensure security and consistency.

By reducing manual effort, preventing impersonation, and improving accuracy, the proposed system provides a modern, efficient, secure, and reliable attendance management solution. It saves time, enhances classroom productivity, and supports the digital transformation of educational institutions through intelligent and automated attendance tracking.

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Note: All figures, user interface screenshots, diagrams, and visual materials presented in this report are available in high quality in the project's GitHub repository for reference, verification, and future use. GitHub Repository: <https://github.com/Diiezel/Smart-Face-Recognition-Attendance-System>

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List of abbreviations and acronyms

API	Application Programming Interface
CS	Computer Science
CSV	Comma-Separated Values
DB	Database
ERD	Entity Relationship Diagram
FRS	Face Recognition System
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
MVC	Model-View-Controller
MySQL	My Structured Query Language
QR	Quick Response
REST	Representational State Transfer
UI	User Interface
UML	Unified Modeling Language
URL	Uniform Resource Locator
JWT	JSON Web Token
TA	Teaching Assistant
CRUD	Create, Read, Update, Delete
RN	React Native

Chapter 1: Introduction

1.1 Overview

In recent years, educational institutions have increasingly adopted digital solutions to improve academic and administrative processes. One of the most essential daily academic activities is attendance tracking, as it directly affects student evaluation, institutional reporting, and academic compliance.

Traditional attendance methods, such as paper-based records, manual calling, and card-based systems, are inefficient and vulnerable to human error and manipulation. These approaches consume valuable lecture time and may allow proxy attendance, which negatively impacts data accuracy and fairness.

To overcome these limitations, this project proposes a **Smart Face Recognition Attendance System** that utilizes **Face Recognition** technology and **QR Code verification**. The system aims to provide a fast, secure, and accurate method for recording attendance while reducing administrative workload, preventing fraudulent attendance, and enhancing transparency within educational institutions. Furthermore, it offers automated attendance management and reporting features that improve efficiency and support the digital transformation of modern educational environments.

1.2 Motivation

The motivation behind this project arises from the growing need to modernize attendance management in educational environments. Manual and semi-digital systems are no longer suitable for institutions with large student populations and multiple academic activities running simultaneously.

Attendance fraud, such as students marking attendance on behalf of absent peers, remains a major concern. In addition, academic staff often face difficulties in organizing attendance data, generating accurate reports, and maintaining consistency across courses. These challenges highlight the importance of developing an automated

system that ensures reliable identity verification, minimizes errors, saves time during lectures, and provides efficient attendance management and reporting.

1.3 Objective

The main objectives of this project are as follows:

- Automate the attendance-taking process using facial recognition and QR code technology.
 - Prevent proxy and fraudulent attendance attempts.
 - Reduce the time required to record attendance during lectures and academic sessions.
 - Ensure secure, accurate, and real-time attendance recording.
 - Provide students with access to their enrolled courses, attendance history, and attendance percentage.
 - Enable academic staff to generate reliable and detailed attendance reports.
 - Support scalability for future system enhancements.
 - Support role-based access control for students and doctors.
-

1.4 Aim

The primary aim of this project is to design and implement an intelligent mobile-based Attendance Management System that ensures accuracy, security, and efficiency in attendance tracking. By integrating **Face Recognition** and **QR Code** verification within a unified platform, the system seeks to enhance fairness, reduce administrative burdens, and improve the overall academic workflow.

1.5 Scope

The scope of the proposed system includes the following functionalities:

- User authentication for students and academic staff.
- Facial data enrollment for students through **image upload or in-app camera capture**.
- Viewing available courses and enrolling in selected courses by students.
- QR code generation for each student.
- Attendance session creation, opening, and closing by academic staff.
- Attendance recording using **Face Recognition** or **QR Code** scanning.
- Secure storage of attendance records in a centralized database.
- Viewing attendance history and attendance percentage by students.
- Generating detailed attendance reports for academic staff.
- Secure backend communication and data management.

1.6 General Constraints

The system may face several general constraints, including:

- Dependence on stable internet connectivity for real-time verification.
- Variations in camera quality across different mobile devices.
- Environmental factors such as lighting conditions affecting facial recognition accuracy.
- Device compatibility limitations across different operating systems.
- Privacy and data protection requirements related to biometric data.
- Possible inaccuracies in facial recognition under extreme conditions.

1.7 Organization of the Project

This project report is organized as follows:

Chapter 1 introduces the project, including the overview, motivation, objectives, scope, and constraints.

Chapter 2 presents the background and previous work.

Chapter 3 covers planning and analysis, including the feasibility study, system analysis, and requirements.

Chapter 4 discusses the system design, including UML diagrams and database design.

Chapter 5 presents the implementation details, software architecture, user interface, and results.

Chapter 6 presents the testing process, including unit testing and integration testing.

Finally, the report concludes with the conclusion, future work, references, and appendices.

Chapter 2: Background and Previous work

2.1 Background

In recent years, the integration of smart technologies and artificial intelligence has transformed many aspects of daily life, including education. Attendance management is one of the most essential administrative processes in academic institutions, as it directly affects student performance evaluation, participation tracking, and institutional reporting. However, traditional attendance methods such as manual sign-in sheets or verbal roll calls remain time-consuming, error-prone, and vulnerable to proxy attendance.

To overcome these challenges, biometric technologies have become a popular solution. Among them, **Face Recognition** is one of the most effective and user-friendly methods, as it provides contactless, fast, and accurate identification. It eliminates the need for physical cards or manual verification while ensuring integrity and efficiency.

The proposed **Smart Face Recognition Attendance System** integrates **Face Recognition** and **QR Code** verification to automatically record attendance during lectures and lab sessions. This dual-method approach improves both accuracy and reliability while reducing the risk of fraud and human error. The system utilizes **React Native** for mobile application development, **RESTful APIs** for backend communication, and **MySQL** for secure and efficient data management. In addition, it supports course enrollment, attendance session management, role-based access control, and attendance reporting, contributing to the digital transformation of educational institutions and providing a smarter and more efficient academic environment.

2.2 Previous Work

The automation of attendance management has been widely explored in recent years, with various biometric and non-biometric techniques developed to improve accuracy,

convenience, and efficiency. Early systems primarily aimed to replace manual roll calls and paper-based records, but each faced specific limitations.

Early Systems

RFID-based systems:

Students were issued RFID cards scanned by readers to mark attendance. These systems reduced paperwork and saved time but were vulnerable to manipulation, such as card sharing, and required dedicated hardware in each classroom.

Fingerprint recognition:

Fingerprint-based systems provided higher accuracy due to the uniqueness of biometric data. However, they required physical contact with the scanner, which raised hygiene concerns and slowed down attendance in large classes.

Face Recognition Evolution

Face recognition began with feature-based methods such as **Eigenfaces** and **Fisherfaces**, which extracted facial features for identification. These techniques were limited in real-world scenarios due to variations in lighting, pose, and facial expressions.

The introduction of deep learning, particularly **Convolutional Neural Networks (CNNs)**, significantly enhanced face recognition performance. Modern models include:

- **DeepFace (2014):** Achieved near-human accuracy using a deep neural network trained on millions of facial images.
- **FaceNet (2015):** Generated 128-dimensional facial embeddings for highly accurate and real-time identification.
- **OpenFace and DeepID:** Open-source implementations that enabled practical academic and industrial applications.

Hybrid Systems

To improve reliability, hybrid systems combined face recognition with other methods such as **QR Codes**, **NFC**, mobile verification, and **GPS** tracking. For example, some universities required students to display a time-sensitive QR code through a mobile application as an alternative attendance verification method. These approaches increased flexibility and accuracy but often lacked a fully integrated mobile platform or comprehensive reporting capabilities.

The proposed **Smart Face Recognition Attendance System** builds upon these approaches by integrating **Face Recognition** and **QR Code** verification within a unified mobile platform supported by **RESTful APIs** and centralized database management. In addition, the system provides course enrollment, attendance session management, and detailed reporting features, offering a secure, efficient, and scalable solution for modern educational institutions.

Limitations of Previous Systems:

Despite partial successes, previous attendance systems had several shortcomings:

1. Lack of real-time synchronization between devices and servers.
2. Limited or unintuitive user interfaces, often desktop-only.
3. Insufficient reporting and visualization tools for attendance analysis.
4. Security vulnerabilities, including replay attacks and face spoofing.

Most prior research focused on either face recognition or QR-based systems individually, rather than combining both methods for dual authentication.

Contribution of the Proposed System:

The proposed **Smart Face Recognition Attendance System** addresses these gaps by integrating AI-based face recognition with QR code verification within a cross-platform mobile application. Key features include:

- Contactless, fast identification using deep learning models.
- Backup verification via dynamically generated QR codes.
- Secure data management with encryption, authentication tokens, and protection against replay attacks.
- Comprehensive analytics and reporting for academic staff.
- Responsive design via Flutter for Android and iOS devices.

Compared to earlier work, this system offers a complete, secure, and user-friendly solution, enhancing efficiency and supporting digital transformation in educational institutions.

Chapter 3: Planning and Analysis

3.1 Planning

Planning is a critical phase in the development of any successful software system. It defines how the project will be executed, managed, and delivered within the specified time and available resources. For the Smart Face Recognition Attendance System, careful planning was carried out to ensure feasibility, accuracy, and alignment with the project goals. This phase includes identifying the required hardware and software resources, defining system features, estimating implementation effort, and scheduling development tasks to achieve a clear and structured workflow throughout the project.

3.1.1 Feasibility study and estimated cost

A feasibility study was conducted to evaluate the practicality of implementing the Smart Face Recognition Attendance System. The study focused on three main aspects:

Technical Feasibility:

The system is technically feasible because it relies on widely used and well-supported technologies such as **React Native** for mobile application development, the **Laravel** framework for backend services, and **MySQL** as the relational database management system. In addition, the face recognition functionality is implemented using a reliable **Face Recognition API**, providing accurate and secure facial verification. These technologies are stable, scalable, and suitable for developing a real-time attendance management system, and the development team possesses the necessary technical skills to implement them successfully.

Economic Feasibility:

The system is economically feasible since it mainly depends on open-source technologies such as **React Native**, **Laravel**, and **MySQL**, significantly reducing development costs. No specialized hardware is required because attendance can be recorded using standard smartphones equipped with built-in cameras. Furthermore,

hosting and deployment can be achieved through low-cost cloud services, making the overall project cost minimal.

Operational Feasibility:

The system is operationally feasible because it is designed with a simple and user-friendly interface. Students can easily register their facial data, enroll in courses, and display their personal QR codes, while doctors can create attendance sessions, verify attendance using **Face Recognition** or **QR Code** scanning, and generate attendance reports efficiently. Therefore, the system can be adopted in real educational environments without requiring advanced technical knowledge.

3.1.2 Gantt chart

A Gantt chart was prepared to represent the project schedule and track progress throughout the development lifecycle of the Smart Face Recognition Attendance System. The chart divides the project into sequential phases, including requirements analysis, system design, database design, frontend development, backend development, integration, testing, documentation, and deployment. Each phase is assigned a specific duration to ensure timely completion of the project and proper task management.

	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9
Requirements Analysis									
System Design									
Database Design									
Frontend Development									
Backend Development									
Integration									
Testing									
Documentation									
Deployment									

Table 3.1 Gantt chart

3.2 Analysis and limitations of existing system

Traditional attendance systems in universities still rely heavily on manual methods such as paper sheets, verbal roll-calling, or basic QR-based solutions without strong verification. These approaches are often slow, prone to human error, and make it easy for students to commit proxy attendance. In addition, managing attendance records manually becomes difficult when dealing with large classes, multiple courses, and frequent sessions, leading to inconsistent tracking and delayed reporting.

3.3 Need for new system

The need for a new attendance system arises from the increasing demand for a faster, more accurate, and more secure way to record attendance in academic environments. The proposed system provides an automated solution that allows academic staff to take attendance using either **Face Recognition** or **QR Code** scanning through a mobile application. This approach reduces time wasted during lectures, prevents cheating, improves record accuracy, and stores all attendance data digitally for reporting and future analysis. In addition, it enables students to monitor their attendance statistics while allowing academic staff to generate detailed attendance reports efficiently.

3.4 Analysis of new system

The proposed system is a mobile-based smart attendance management solution designed for universities and academic institutions. It provides a centralized platform where students and academic staff can manage courses, attendance sessions, and attendance records in a fully digital environment. Students can register their facial data, enroll in available courses, generate their personal QR codes, and view their attendance reports and percentages. Academic staff can create courses, manage attendance sessions, verify attendance using **Face Recognition** or **QR Code** scanning, and generate detailed attendance reports.

The system focuses on speed, accuracy, and security by reducing proxy attendance through biometric verification and QR-based authentication. All attendance data is stored in a structured database with role-based access control, enabling efficient reporting, tracking, and future system expansion.

3.4.1 User Requirements

1) Student Requirements

- Students must be able to create an account and log in.
- Students must be able to register their face using the app camera or image upload.
- Students must be able to view available courses and enroll in selected courses.
- Students must be able to view their attendance history and attendance percentage.
- Students must be able to show their personal QR code to academic staff during attendance.

2) Staff Requirements

- Staff must be able to log in and access their dashboard.
- Staff must be able to create, start, and close attendance sessions.
- Staff must be able to create and manage courses.
- Staff must be able to take attendance using Face Recognition.
- Staff must be able to take attendance using QR Code scanning.
- Staff must be able to generate and export attendance reports.
- Staff must be able to view attendance results for each session.

3) General User Requirements

- The application must be easy to use for both students and academic staff.
- Error messages must be clear and helpful.
- The application must load quickly and respond smoothly.

- Users must be notified of failed operations such as login failure, invalid face verification, or QR scanning errors.
-

3.4.2 System Requirements

1) System Overview

- The system must provide attendance tracking using Face Recognition and QR Code scanning.
- The system must allow students and academic staff to log in and access their dashboards.
- The system must manage courses, enrollments, attendance sessions, and attendance records.

2) Backend Requirements

- The backend must expose secure RESTful APIs.
- The backend must handle Face Recognition requests.
- The backend must manage user authentication using secure tokens.
- The backend must store attendance records, face tokens, enrollment data, and course information.

3) Database Requirements

- The database must store users, courses, enrollments, attendance sessions, and attendance records.
- The database must ensure data integrity using primary and foreign key constraints.

4) Face Recognition Requirements

- The system must process student face images and generate a secure face token.
- The system must compare attendance face scans against stored face tokens.
- The system must reject unclear or low-quality face images.

5) QR Code Requirements

- The system must generate a unique QR Code for each student.
- The system must validate QR Codes quickly during scanning.
- The system must record attendance based on QR Code verification.
- The system must prevent duplicate attendance records within the same attendance session.

6) Attendance Session Requirements

- Academic staff must be able to create, open, and close attendance sessions.
- The system must not allow attendance recording for closed sessions.
- The system must support attendance through either Face Recognition or QR Code scanning.

7) Reporting Requirements

- The system must generate attendance reports for academic staff.
 - Reports must include attendance totals, percentages, and student attendance history.
 - Students must be able to view their attendance percentage for each enrolled course and their overall attendance statistics.
-

3.4.3 Domain Requirements

The system operates within the university education domain and focuses on managing and automating student attendance. It supports course enrollment, attendance session creation, and attendance recording using Face Recognition or QR Code scanning while securely storing attendance data for monitoring, reporting, and analysis by academic staff.

3.4.4 Functional Requirements

1) User Management

- Users (students and academic staff) must be able to create accounts.
- Users must be able to log in using email and password.

- The system must validate login credentials.
- The system must assign permissions based on the user's role.

2) Face Enrollment

- The system must capture or upload a student's face image.
- The image must be sent to the backend for processing.
- The system must validate the quality of the face image.
- The system must generate and store a unique face token.
- The system must confirm successful face registration.

3) Course Management

- The system must display available courses.
- Students must be able to enroll in selected courses.
- The system must store enrolled courses for each student.

4) Attendance Session Management

- Academic staff must be able to create attendance sessions.
- Academic staff must select the attendance method (Face Recognition or QR Code).
- The system must allow opening and closing attendance sessions.

5) Face Attendance

- Academic staff must be able to open Face Attendance mode.
- The system must capture and send the student's face image to the server.
- The server must compare the face with the stored face token.
- The system must mark the student as present if verification succeeds.
- The system must return the verification result.

6) QR Attendance

- Academic staff must be able to scan student QR Codes.
- The system must validate the QR Code.

- The system must record attendance if the QR Code is valid.
- The system must prevent duplicate attendance for the same student within the same session.

7) Attendance Records

Each attendance record must include:

- user_id
- attendance_session_id
- method
- status
- match_score (for Face Recognition)
- attended_at

8) Attendance Reports

- Academic staff must be able to generate attendance reports.
 - Academic staff must select the required course.
 - The system must display attendance summaries and statistics.
 - Students must be able to view their attendance percentage and attendance history.
-

3.4.5 Non- Functional Requirements

1) Performance

- Face Recognition must complete within 2–3 seconds.
- QR Code attendance must be processed within 1 second.
- The backend must support multiple concurrent attendance requests.

2) Reliability

- The system must maintain high availability during operation.
- Attendance data must never be lost.
- If the Face Recognition service is unavailable, the system should continue operating by allowing QR Code attendance.

3) Security

- Passwords must be securely encrypted using hashing algorithms such as bcrypt.
- All communication between the mobile application and the backend must use HTTPS.
- Students' facial data and face tokens must be securely protected and never exposed.
- Token-based authentication must be implemented using Laravel Sanctum.

4) Usability

- The system must be easy for both students and academic staff to use.
- Face registration steps must be simple and clear.
- Error messages must be understandable and informative.

5) Accuracy

- Face Recognition confidence should be 90% or higher for successful verification.
- The system must reject low-quality or invalid face images.
- The system must prevent incorrect or duplicate attendance records.

6) Scalability

- The system must support thousands of students.
- The system must handle future growth in courses and attendance sessions.

7) Maintainability

- The source code must be modular and easy to maintain.

- The database structure must support future extensions without major modifications.

8) Availability

- The system must be available during all academic working hours.
- System recovery after unexpected failures should be fast and reliable.

9) Backup and Recovery

- Regular backups of user and attendance data should be performed.
- The system should be able to restore data quickly in case of failure.

10) Compatibility

- The mobile application must work on both Android and iOS devices.
 - The backend REST APIs must be fully compatible with React Native applications.
-

3.5 Advantages of new system

- Faster attendance recording compared to manual methods.
 - Higher accuracy by preventing proxy attendance.
 - Flexible attendance using either Face Recognition or QR Code scanning.
 - Secure storage of attendance records for tracking and reporting.
 - Low cost and easy deployment using standard mobile devices.
-

3.6 User characteristics

Students

Students are the primary users responsible for registering their facial data, enrolling in courses, generating their QR Codes, and viewing their attendance history and attendance percentages. The application should be simple and easy to use without requiring technical knowledge.

Academic Staff (Doctors / TAs)

Academic staff are responsible for creating and managing courses, creating attendance sessions, selecting the attendance method (Face Recognition or QR Code), recording student attendance, generating and exporting reports, and monitoring attendance statistics. The interface should be fast, intuitive, and efficient during lectures.

Chapter 4: Design

4.1 Design and Implementation Constraints

The design and implementation of the proposed **Smart Face Recognition Attendance System**

were carried out while considering several technical and operational constraints. These constraints influenced the system architecture, database design, and the selection of development tools and technologies.

One of the main constraints was the need to develop a secure and reliable attendance system that could support different attendance methods, including **Face Recognition** and **QR Code** verification. The system had to ensure accurate identification of students while preventing duplicate or fraudulent attendance records.

Another important constraint was the compatibility between the backend services and the mobile application. Therefore, the system was designed using a **RESTful API architecture**, allowing seamless communication between the Laravel backend and the React Native mobile application through standardized HTTP requests and JSON responses.

The implementation was also constrained by the availability and limitations of third-party services. The face recognition functionality depends on an external API, which requires internet connectivity and is subject to usage limitations and response times. Consequently, the system was designed to handle API responses efficiently and provide appropriate error handling when necessary.

Database consistency and integrity represented another critical constraint. The database was designed using a relational model with primary and foreign keys to ensure accurate relationships between users, courses, enrollments, attendance sessions, and attendance records while minimizing data redundancy.

Security considerations also played a significant role in the system design. User authentication, role-based access control, and secure API communication were

implemented to ensure that only authorized users can access the functionalities assigned to their roles.

Finally, the system was designed with scalability and maintainability in mind. The modular architecture allows future enhancements, such as supporting additional attendance methods, generating more advanced reports, or integrating with university management systems, without requiring major changes to the existing implementation.

4.2 Assumptions and Dependencies

The design and implementation of the proposed **Smart Face Recognition Attendance System** are based on several assumptions and dependencies that ensure the proper operation of the system and its components. These assumptions define the expected environment in which the system will function, while the dependencies represent the technologies and external services required for successful execution.

It is assumed that every user has a valid account within the system and is assigned an appropriate role, either as a **student** or a **doctor**. Students are expected to register their personal information and complete the face registration process before using the face recognition attendance feature. Similarly, doctors are assumed to create courses and attendance sessions before students can record their attendance.

The system also assumes that students enroll only in the courses available to them and that attendance is recorded during an active attendance session. Once a session is closed, no additional attendance records can be added for that session.

From a technical perspective, the system depends on a stable internet connection to enable communication between the mobile application and the backend server. Since face recognition functionality utilizes an external AI service, internet connectivity is essential for processing and verifying facial images.

The backend of the system is implemented using the **Laravel Framework**, while the mobile application is developed using **React Native**. The system also relies on a **MySQL relational database** to store users, courses, enrollments, attendance sessions, and attendance records while maintaining data integrity through relational constraints.

Furthermore, the QR Code attendance mechanism depends on the ability of the mobile device to generate and scan QR codes correctly. The face recognition feature depends on the availability and proper operation of the external Face Recognition API service.

Finally, it is assumed that users interact with the system through supported devices and modern operating systems capable of running the mobile application and accessing the required hardware components such as the camera for face registration and QR code scanning.

4.3 Risks and Risk Management

The development and deployment of the proposed **Smart Face Recognition Attendance System** involve several technical and operational risks that may affect the system's performance, reliability, and usability. Therefore, appropriate risk management strategies were considered during the design and implementation phases to minimize their impact.

One of the primary risks is the dependency on external face recognition services. Since the face verification process relies on a third-party API, any service interruption, network failure, or API limitation may temporarily affect the attendance process. To mitigate this risk, the system includes proper error handling and allows the use of an alternative attendance method through **QR Code** verification.

Another potential risk is internet connectivity. The system depends on communication between the mobile application and the backend server using RESTful APIs. Poor or unstable network connections may delay requests or prevent attendance from being

recorded. The application is designed to notify users of communication failures and allow them to retry their requests.

Security risks are also an important consideration. Unauthorized access, token theft, or malicious API requests could compromise the system. To reduce these risks, the system implements secure user authentication, role-based access control, and protected API endpoints, ensuring that only authorized users can access specific functionalities.

Data integrity represents another possible risk. Duplicate attendance records or invalid enrollments could produce inaccurate reports. To address this issue, the database design enforces relational constraints and validation rules that prevent duplicate attendance registration and ensure that only students enrolled in a course can record attendance for its sessions.

Operational risks such as incorrect face registration or damaged QR codes may also occur. The system validates uploaded images during face registration and generates unique QR codes for each student, reducing the possibility of incorrect identification.

Finally, future scalability was considered during system design. As the number of users, courses, and attendance sessions increases, system performance may be affected. Therefore, the backend architecture and relational database were designed to support future optimization and expansion with minimal modifications.

By identifying these risks and implementing appropriate management strategies, the proposed **Smart Face Recognition Attendance System** achieves higher reliability, security, and maintainability while ensuring continuous and accurate attendance management.

4.4 Design of database ERD

The database design is one of the most important components of the proposed **Smart Face Recognition Attendance System**, as it provides a structured way to store, manage, and retrieve all system data efficiently. A well-designed database ensures data integrity, minimizes redundancy, and supports the system's functional requirements.

The Entity Relationship Diagram (ERD) was designed to represent the relationships between the main entities of the system. These entities include **users**, **courses**, **enrollments**, **attendance sessions**, and **attendance records**. Each entity has its own attributes and is connected to other entities through defined relationships that reflect the real-world operations of the attendance management process.

The database follows a relational model, where primary keys and foreign keys are used to maintain consistency and enforce referential integrity between tables. This design enables efficient execution of operations such as student enrollment, course management, attendance recording using QR codes or face recognition, and report generation.

Moreover, the proposed database structure is scalable and flexible, allowing future enhancements such as adding new attendance methods, integrating notification services, or expanding the system to support multiple academic departments without significant modifications to the existing design.

The following sections present the Entity Relationship Diagram and the mapping of its entities into the corresponding relational database tables.

4.4.1 Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrates the logical structure of the database designed for the **Smart Face Recognition Attendance System**. It identifies the main entities within the system and defines the relationships among them to ensure efficient data organization and management.

The proposed ERD consists of several core entities, including **Users**, **Courses**, **Enrollments**, **Attendance Sessions**, and **Attendance Records**. Each entity represents a specific component of the system and stores information required to support the attendance management process.

The **Users** entity stores information about both students and doctors, while the **Courses** entity contains the details of academic courses created by doctors. Since a student can register for multiple courses and each course can have multiple students, the **Enrollments** entity is used to implement the many-to-many relationship between students and courses.

For attendance management, each course can have multiple **Attendance Sessions**, where every session represents a lecture or class meeting. The **Attendance Records** entity stores the attendance status of students for each session and records the attendance method, such as **Face Recognition** or **QR Code**.

The ERD was designed following the principles of database normalization to minimize data redundancy and maintain data consistency. Primary keys and foreign keys are utilized to establish relationships between entities and enforce referential integrity throughout the database.

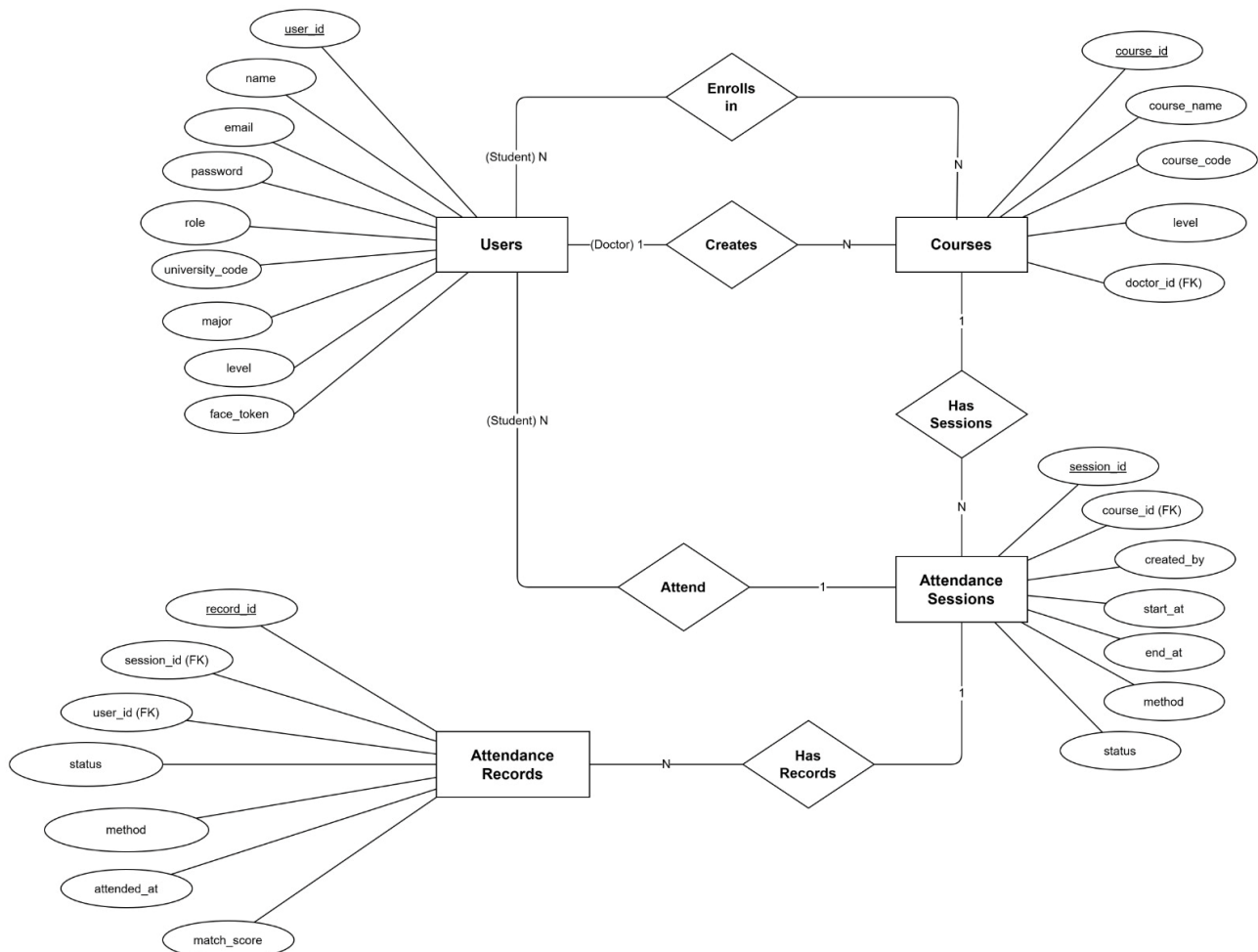


Figure 4.1 ERD

4.4.2 Mapping of Entity Relationship Diagram

The Entity Relationship Diagram (ERD) was transformed into a relational database schema by mapping each entity into a corresponding database table. This process ensures that the conceptual design is converted into a practical implementation that can be managed efficiently by the Database Management System (DBMS).

Each entity in the ERD was mapped to a separate table containing its attributes as columns. Every table has a **Primary Key (PK)** that uniquely identifies each record, while **Foreign Keys (FK)** are used to establish relationships between related tables and maintain referential integrity.

The **Users** entity was mapped into the **Users** table, which stores information about both students and doctors, including personal details and authentication data. The **Courses** entity was mapped into the **Courses** table, where each course is associated with its corresponding doctor through the **doctor_id** foreign key.

Since a many-to-many relationship exists between students and courses, an intermediate table called **Enrollments** was created. This table contains references to both the **Users** and **Courses** tables, allowing each student to enroll in multiple courses while each course can have multiple enrolled students.

The **Attendance Sessions** entity was mapped into the **Attendance_Sessions** table, where each session belongs to a specific course and is created by a doctor. Similarly, the **Attendance Records** entity was mapped into the **Attendance_Records** table, which links students to attendance sessions and stores the attendance status and attendance method, such as **Face Recognition** or **QR Code**.

The mapping process follows the principles of relational database normalization, reducing data redundancy while preserving consistency and integrity. This structure enables efficient querying, updating, and reporting, making it suitable for the operational requirements of the **Smart Face Recognition Attendance System**.

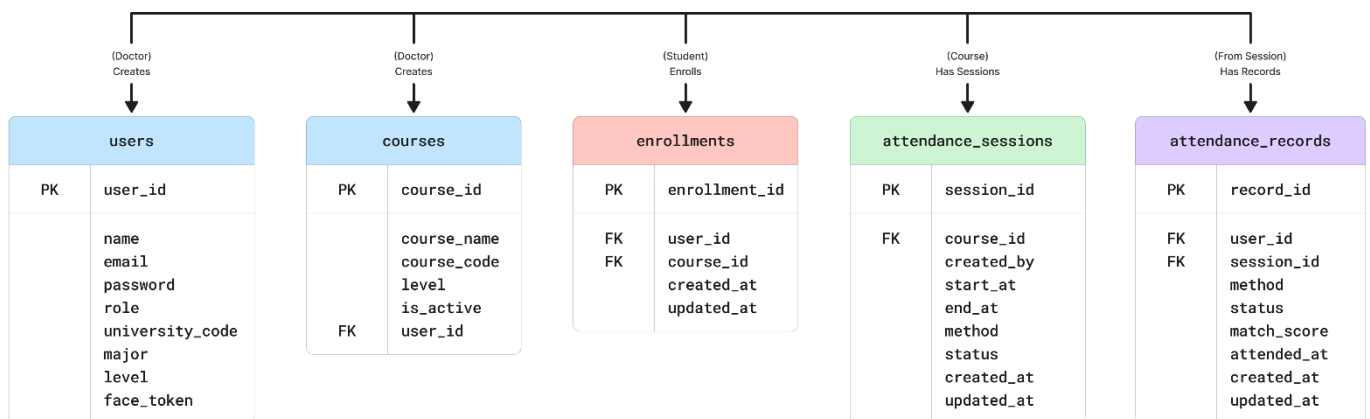


Figure 4. 2 Mapping of ERD

4.5 Class Diagram

The Class Diagram represents the static structure of the **Smart Face Recognition Attendance System** by illustrating the main classes, their attributes, methods, and the relationships between them. It provides an object-oriented view of the system and serves as a blueprint for implementing the software architecture.

The proposed system consists of several core classes that cooperate to perform the required functionalities. The **User** class represents both students and doctors and contains the necessary information related to authentication and user profiles. The **Course** class stores course information and maintains its relationship with the doctor who created it.

The **Enrollment** class acts as an association between students and courses, allowing a many-to-many relationship where one student can enroll in multiple courses and one course can include multiple students. The **AttendanceSession** class represents each attendance session created by a doctor for a specific course and manages the attendance status during that session.

Furthermore, the **AttendanceRecord** class stores the attendance information for each student in a particular session, including the attendance method such as **Face Recognition** or **QR Code** and the attendance status. Additional service classes are responsible for handling face recognition processing, QR code generation and verification, report generation, and authentication operations.

The relationships among these classes are established through associations and dependencies that reflect the actual workflow of the system. This object-oriented

design improves modularity, simplifies maintenance, and facilitates future extensions without significantly affecting the existing implementation.

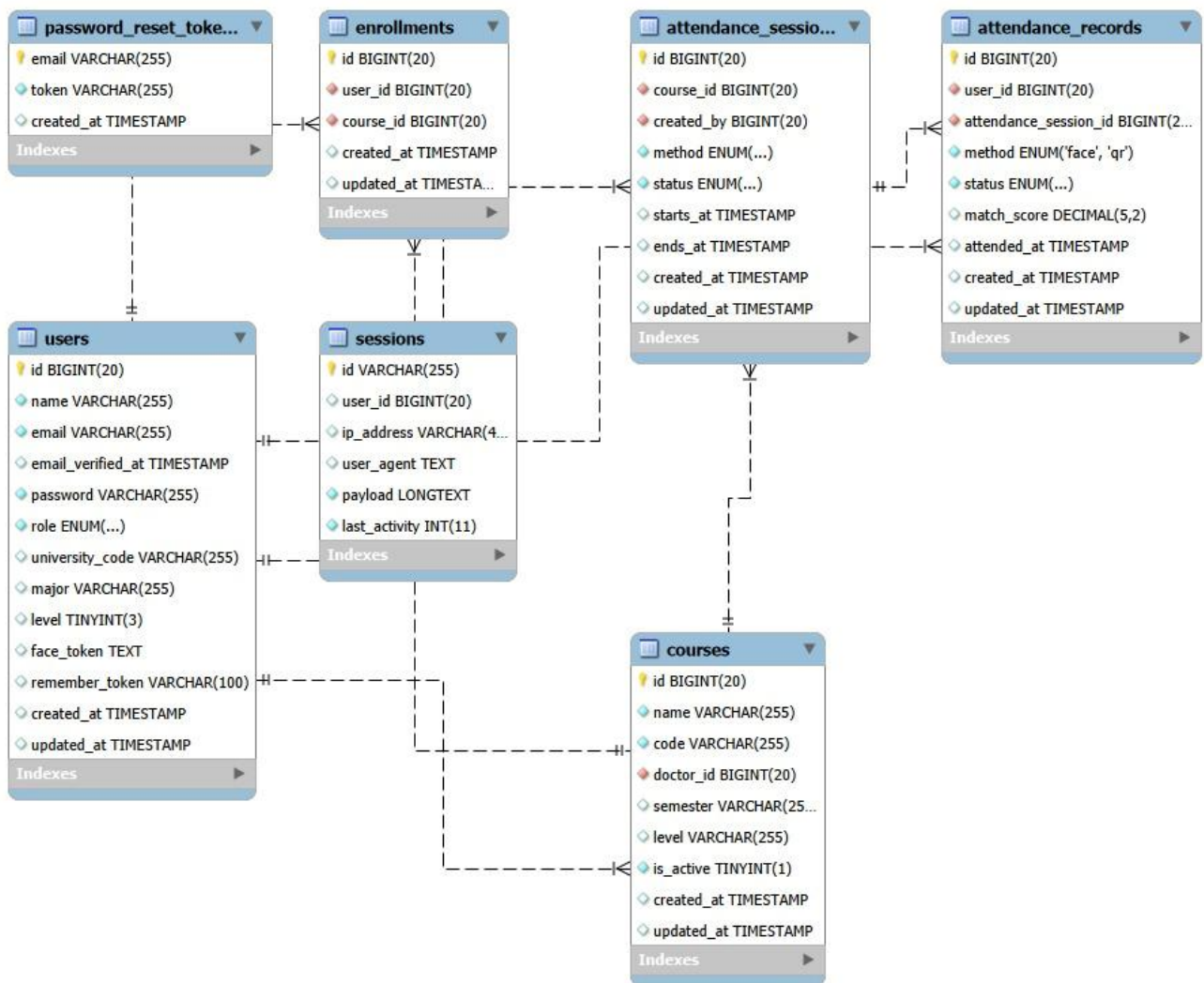


Figure 4.3 Class Diagram

4.6 Use Case Diagram

The Use Case Diagram illustrates the functional requirements of the Smart Face Recognition Attendance System. It shows the interaction between system actors and the available system functionalities. This diagram helps in identifying system behavior from the user's perspective.

4.6.1 Primary Use case Diagram

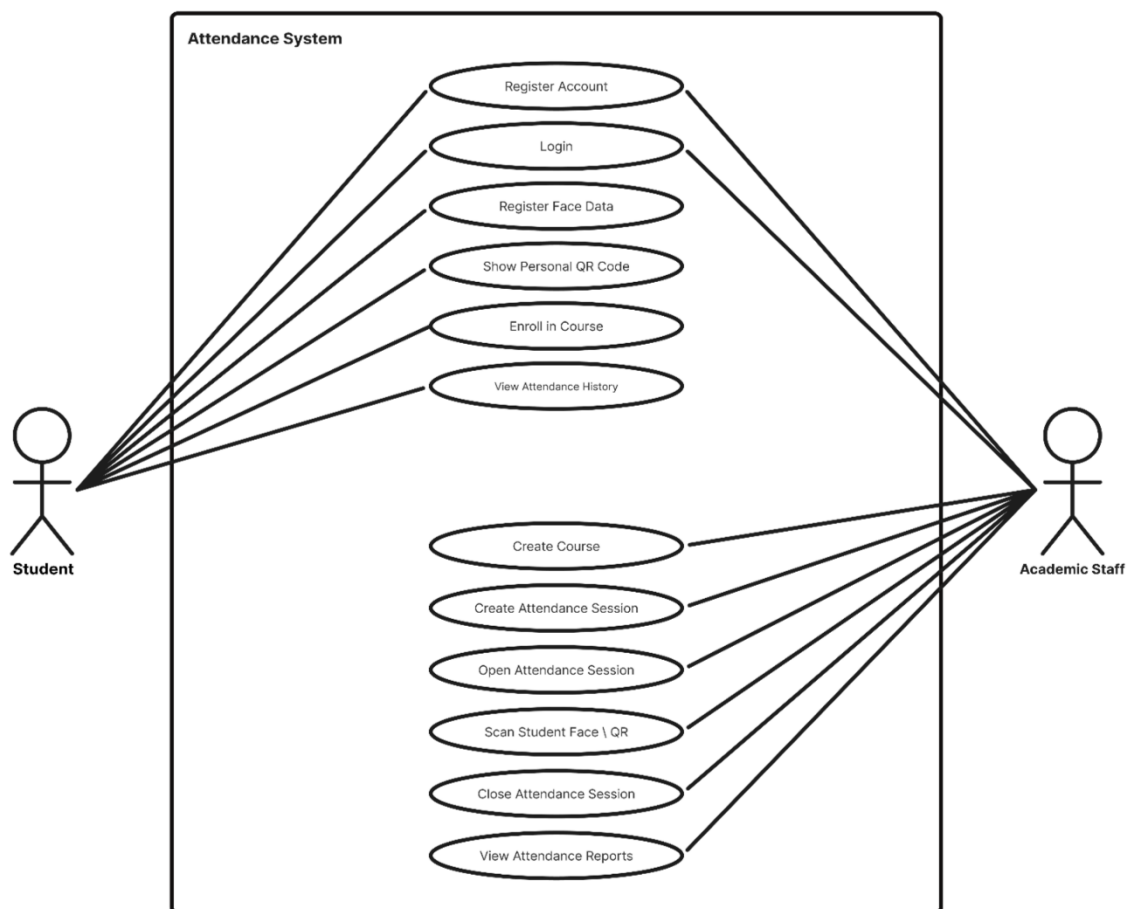


Figure 4.4 Use Cass Diagram

The Use Case Diagram illustrates the interaction between system actors (Admin, Instructor, Student) and the system functions such as login, manage courses, take attendance, and generate reports.

4.6.2 Use-case Scenarios

This section presents the detailed use case scenarios of the Attendance System. Each scenario describes how a specific system function is performed step by step, including the primary actor, preconditions, main flow of events, alternative flows, and postconditions. These scenarios provide a clear understanding of how students and academic staff interact with the system, and how the system handles different situations during the attendance management process.

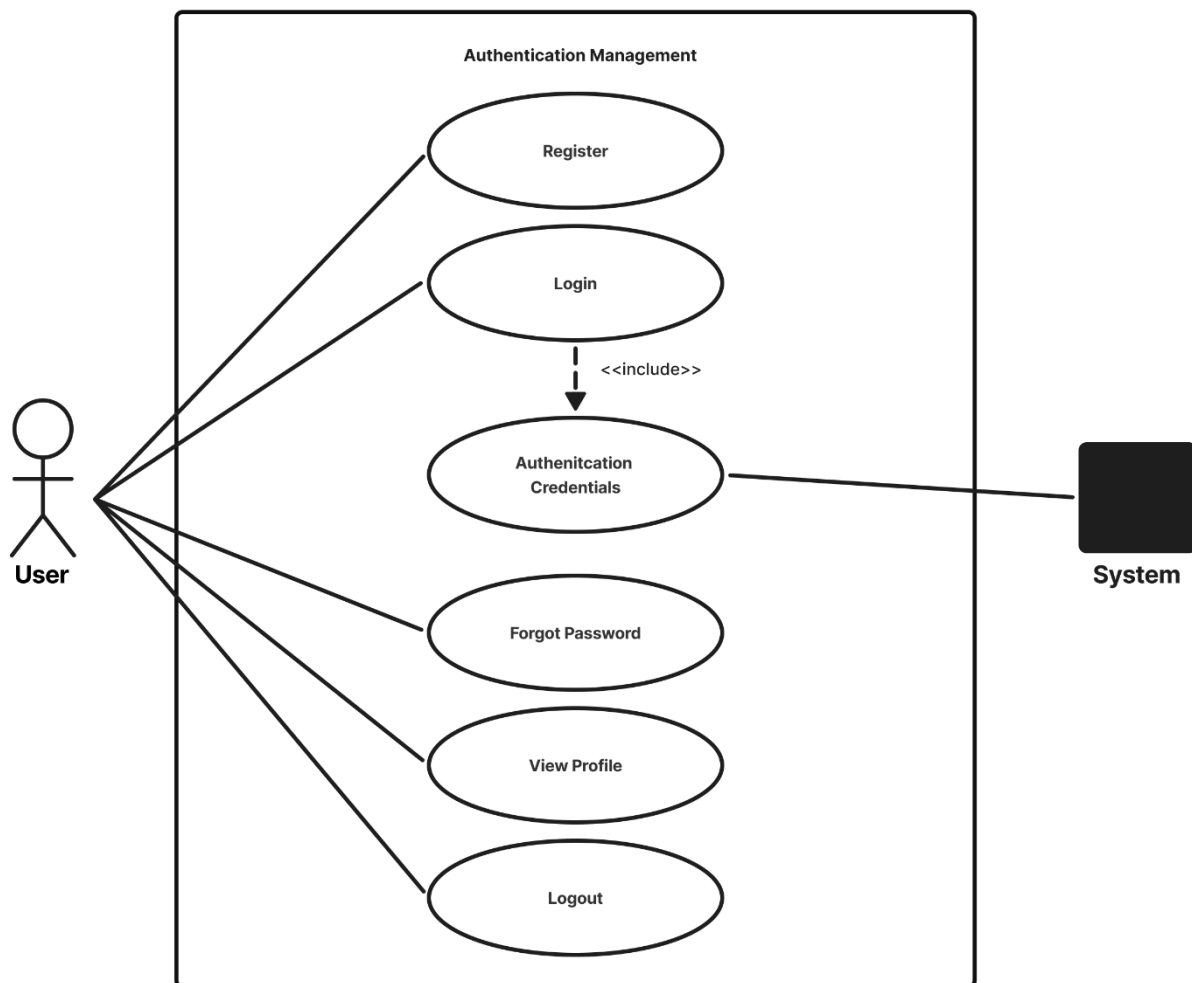


Figure 4. 5 Authentication Management

UC-01: Register Account

Actor(s): Student

Preconditions: System is available.

Main: Student enters registration data :- System validates :- Account is created.

Alternative: Duplicate/invalid data :- System rejects and shows message.

Postconditions: Account is stored.

UC-02: Login

Actor(s): Student / Academic Staff

Preconditions: System is available , Actor is registered.

Main: Actor submits credentials :- System authenticates :- Session starts.

Alternative: Invalid credentials/disabled account :- Access denied.

Postconditions: Actor is logged in.

UC-03: Register Face Data

Actor(s): Student

Preconditions: System is available , Student is logged in.

Main: Student captures face samples :- System checks quality :- Template is stored.

Alternative: Camera denied/poor capture :- Permission or recapture required.

Postconditions: Face template is linked to student profile.

UC-04: Show Personal QR Code

Actor(s): Student

Preconditions: System is available , Student is logged in.

Main: System retrieves/generates student QR code :- Displays it.

Alternative: Generation error :- Retry.

Postconditions: QR code is displayed.

UC-05: Enroll in Course

Actor(s): Student

Preconditions: System is available , Student is logged in.

Main: Student selects course :- System verifies rules :- Student enrolled.

Alternative: Course full/closed or already enrolled :- Enrollment rejected.

Postconditions: Enrollment is saved.

UC-06: View Attendance History

Actor(s): Student

Preconditions: System is available , Student is logged in.

Main: System retrieves attendance records :- Displays them.

Alternative: No records :- 'No records found'.

Postconditions: Attendance history is shown.

UC-07: Create Course

Actor(s): Academic Staff

Preconditions: System is available. , Academic staff is logged in and authorized.

Main: Staff submits course details :- System validates :- Course created.

Alternative: Duplicate course code :- Rejected.

Postconditions: Course is available in the system.

UC-08: Create Attendance Session

Actor(s): Academic Staff

Preconditions: System is available. , Academic staff is logged in and authorized.

Main: Staff defines session details :- System validates :- Session scheduled.

Alternative: Overlapping session :- Rejected.

Postconditions: Session is created.

UC-09: Open Attendance Session

Actor(s): Academic Staff

Preconditions: System is available , Academic staff is logged in and authorized.

Main: Staff opens session :- System sets status to Open :- Accepts check-ins.

Alternative: Session not eligible :- Blocked.

Postconditions: Session is Open.

UC-10: Close Attendance Session

Actor(s): Academic Staff

Preconditions: System is available , Academic staff is logged in and authorized.

Main: Staff closes session :- System sets status to Closed :- Stops check-ins.

Alternative: Session already closed :- No action.

Postconditions: Session is Closed.

UC-11: View Attendance Reports

Actor(s): Academic Staff

Preconditions: System is available , Academic staff is logged in and authorized.

Main: Staff selects course/period :- System generates report :- Displays it.

Alternative: No data :- Empty report message.

Postconditions: Report is generated.

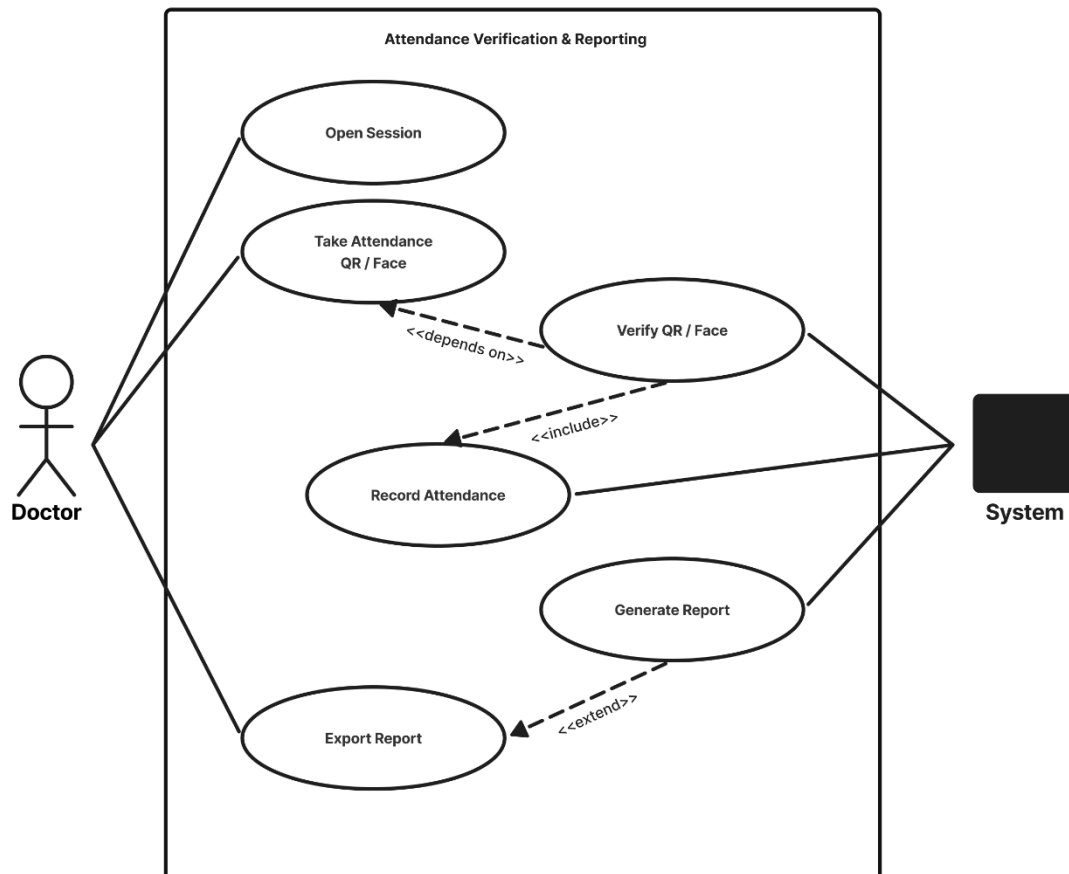


Figure 4.6 Attendance Verification & Reporting

UC-12: Scan Student Face / QR

Actor(s): Student / Academic Staff

Preconditions: System is available , An attendance session is Open.

Main: Actor scans face/QR :- System identifies student :- Attendance recorded.

Alternative: Not recognized/no open session/already checked-in :- Message shown.

Postconditions: Attendance record stored (if successful).

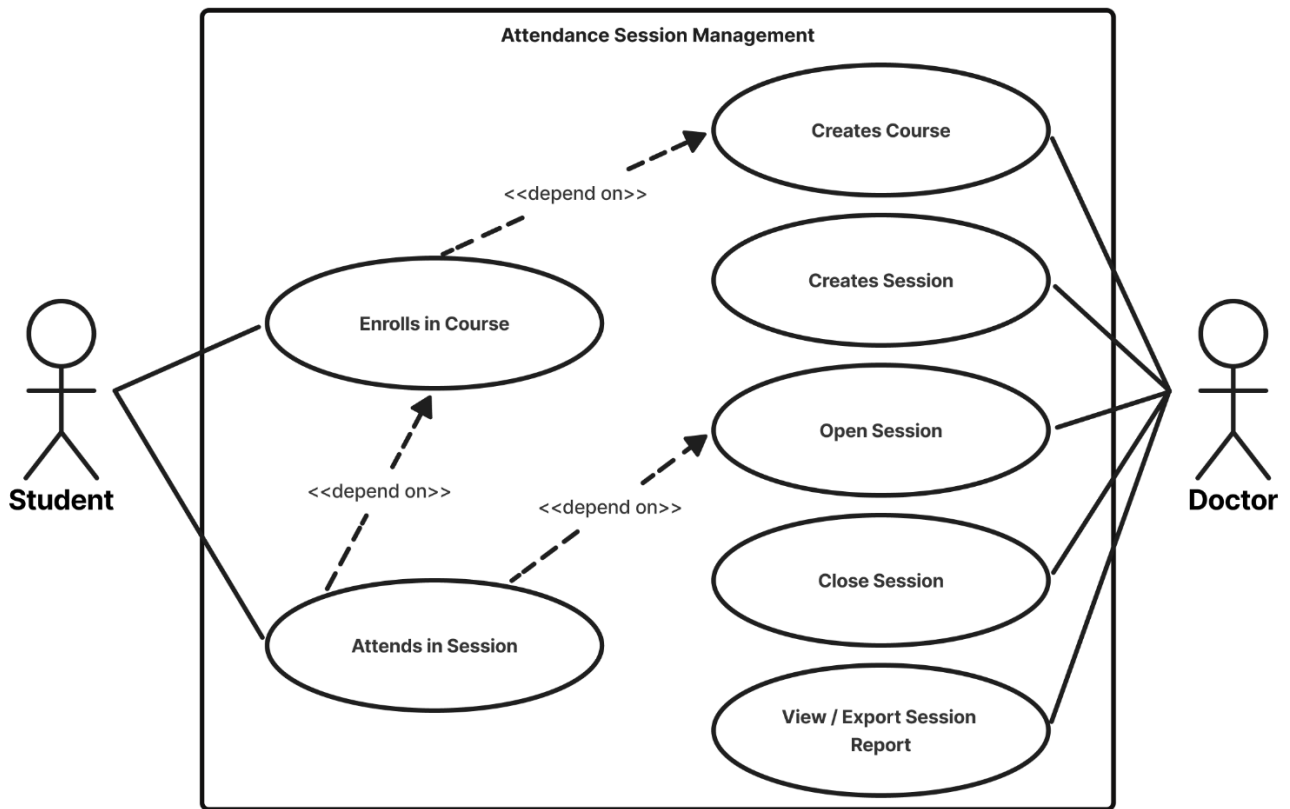


Figure 4.7 Attendance Session Management

4.7 Activity Diagram

The Activity Diagram describes the workflow of the attendance process in the Smart Face Recognition Attendance System. It illustrates the sequence of activities, decision points, and possible user paths, including the student registration process and the staff attendance-taking process using face recognition or QR scanning.

4.7.1 Student Activity Diagram

The diagram shows the student's interaction with the attendance system.

The process starts with login. If successful, the student accesses the dashboard. From there, the student can enroll their face, register courses, or view attendance history. The system validates inputs such as login data and face recognition before confirming actions. The process ends after completing the selected task.

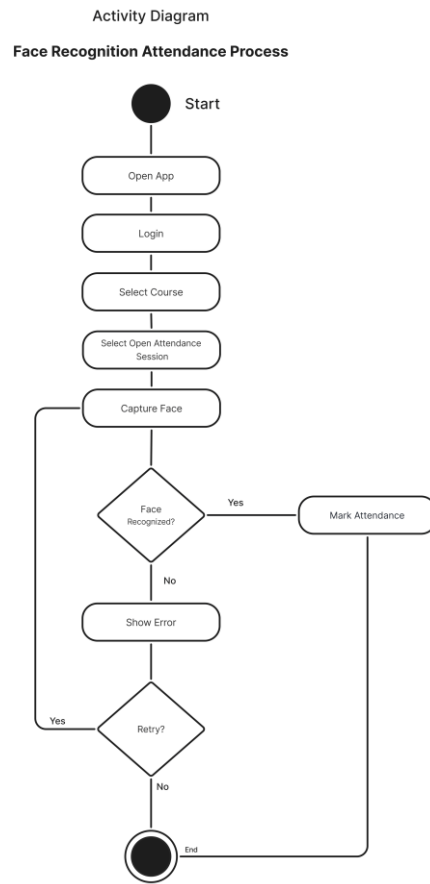


Figure 4.8 Student Activity Diagram

4.7.2 Staff Activity Diagram

The Staff Activity Diagram describes how the instructor uses the attendance system. The process starts with logging into the application. After successful login, the instructor selects a course and opens an attendance session.

Attendance can be recorded using face recognition or QR code scanning. The system validates the face or QR code before marking the student as present. If validation fails, an error message appears.

After completing attendance, the instructor can close the session and generate or export attendance reports. The process ends after finishing these tasks.

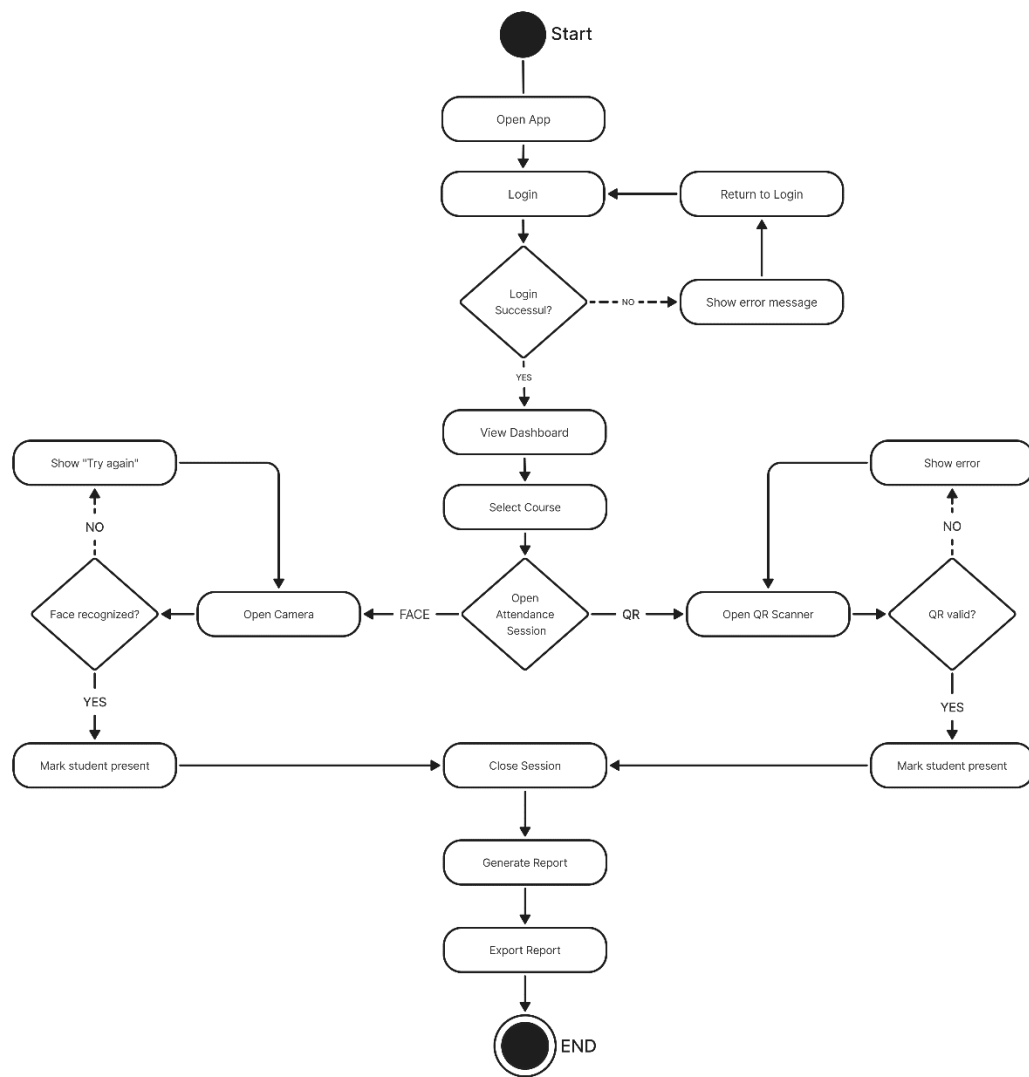


Figure 4.9 Staff Activity Diagram

4.8 Sequence Diagram

The Sequence Diagram illustrates the interaction between different system components over time. It shows how requests and responses are exchanged between

the mobile application, backend server, database, and face recognition service during operations such as login, face enrollment, attendance recording, and report generation.

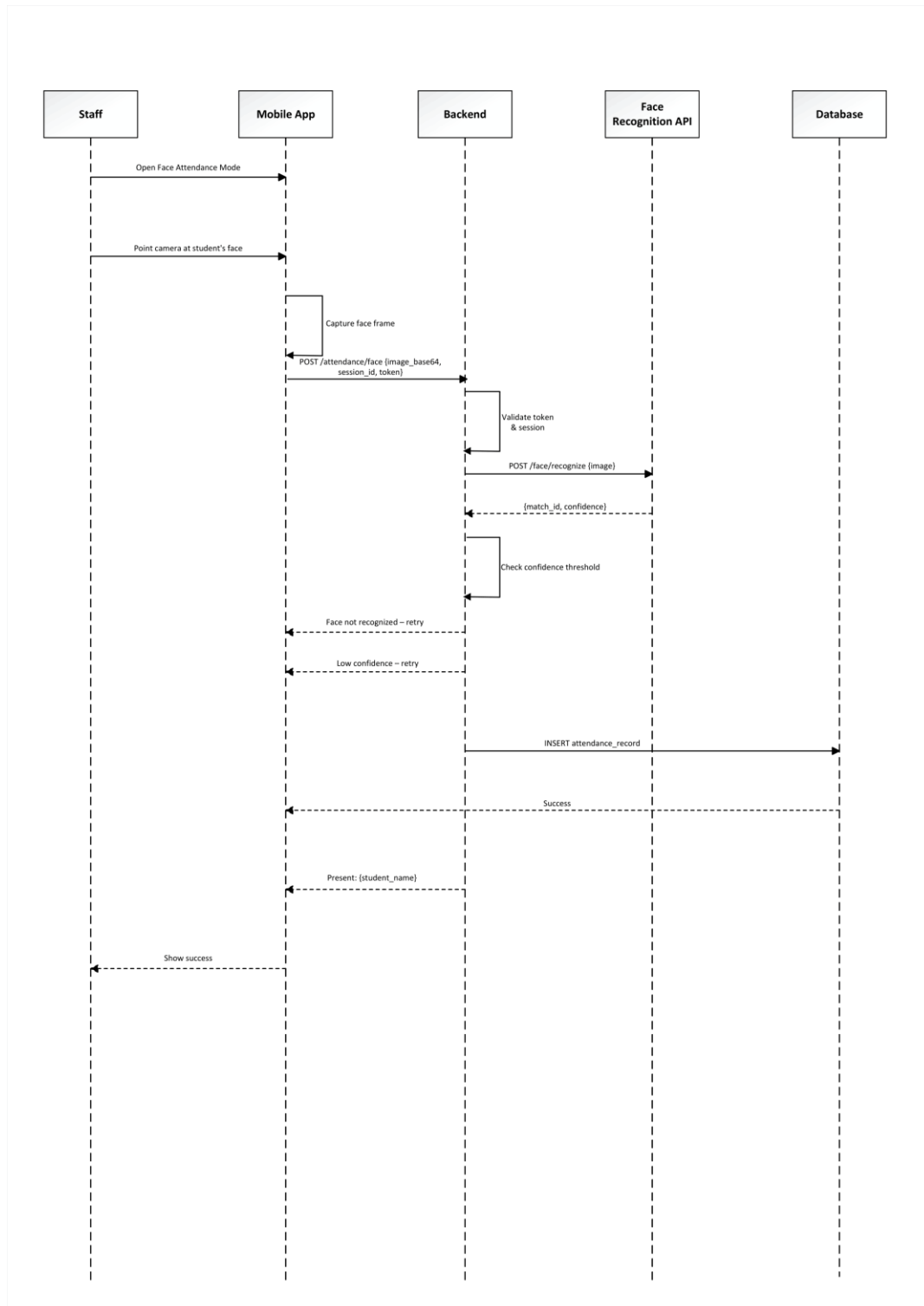


Figure 4.10 Sequence diagram

4.9 State Diagram

The State Diagram illustrates the different states of the **Smart Face Recognition Attendance System** during the attendance process and the transitions between these states based on user actions and system events.

The process begins when a user launches the application and logs in as either a **Student** or a **Doctor**. Students can perform actions such as course enrollment, face registration, QR code generation, and viewing attendance reports. Doctors can create courses, manage attendance sessions, verify attendance using **Face Recognition** or **QR Code**, generate reports, and close sessions.

An attendance session starts in the **Open** state, allowing attendance to be recorded, and changes to the **Closed** state once the doctor ends the session, preventing further modifications. The State Diagram demonstrates these transitions and provides a clear understanding of the system's behavior and workflow.

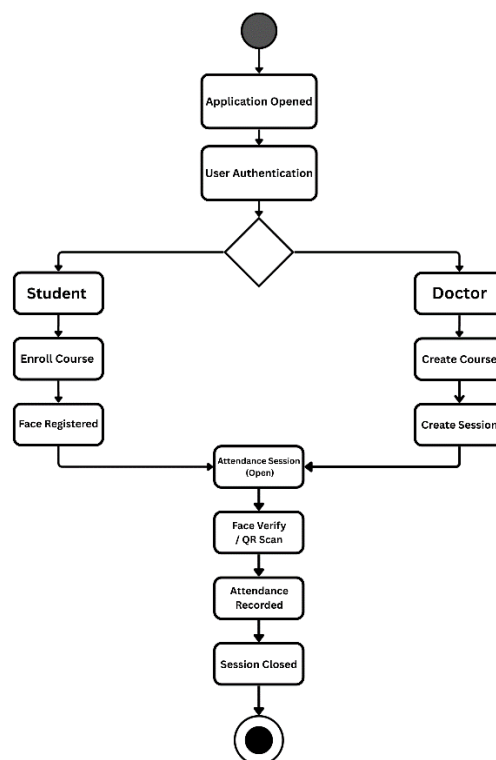


Figure 4.11 State Diagram

Chapter 5: Implementation

5.1 Software Architecture

The proposed **Smart Face Recognition Attendance System** adopts a **client-server architecture**, where the system is divided into two main components: the **mobile application (client)** and the **backend server**. This architecture improves scalability, maintainability, and security by separating the user interface from the business logic and data management.

The **mobile application**, developed using **React Native**, serves as the interface through which students and doctors interact with the system. It allows users to register, log in, enroll in courses, register facial data, generate or scan QR codes, record attendance, and view attendance reports. The application communicates with the backend through secure **RESTful APIs** using the **HTTP protocol**, exchanging data in **JSON format**.

The **backend server** was implemented using the **Laravel Framework**, which is responsible for handling authentication, authorization, business logic, and communication with the database. It processes all requests received from the mobile application, validates user input, executes the required operations, and returns appropriate responses.

The system utilizes a **MySQL relational database** to store and manage all data, including users, courses, enrollments, attendance sessions, and attendance records. Relationships between tables are maintained using primary keys and foreign keys to ensure data consistency and integrity.

To support intelligent attendance verification, the architecture integrates an external **Face Recognition API**, which compares captured facial images with the registered facial data stored for each student. In addition, the system supports **QR Code-based attendance**, where each student is assigned a unique QR code that can be scanned by the doctor during an active attendance session.

Authentication and authorization are implemented using **Laravel Sanctum**, providing secure token-based access to the system's APIs. Furthermore, **role-based access control** ensures that students and doctors can access only the functionalities assigned to their respective roles.

The modular architecture adopted in this project allows each subsystem - authentication, course management, enrollment, attendance management, reporting, and facial recognition - to operate independently while interacting seamlessly with other modules. This design simplifies maintenance, enhances performance, and facilitates future expansion of the system.

5.2 User Interface

The User Interface (UI) of the proposed **Smart Face Recognition Attendance System** was designed to provide a simple, modern, and user-friendly experience for both students and doctors. The interface follows a clean layout with intuitive navigation, allowing users to perform their tasks efficiently with minimal effort.

The user interface was **designed and prototyped using Figma**, which enabled the creation of interactive and organized screen layouts before the development phase. This approach helped visualize the application workflow, improve usability, and ensure a consistent design across all screens.

The mobile application was developed using **React Native**, enabling a responsive interface that works consistently across different mobile devices. The design focuses on accessibility, readability, and ease of use by organizing features into clear screens and menus.

For students, the interface provides functionalities such as account registration, login, face registration, course enrollment, QR code generation, viewing enrolled courses,

and accessing attendance reports. These features are presented through straightforward navigation to ensure a smooth user experience.

For doctors, the interface includes options to create courses, manage attendance sessions, verify student attendance using **Face Recognition** or **QR Code**, monitor attendance records, and generate attendance reports. The design allows doctors to perform these operations quickly during lectures.

Overall, the user interface was designed in **Figma** and implemented using **React Native** to enhance usability, reduce complexity, and provide an efficient interaction between users and the **Smart Face Recognition Attendance System**.

The following figures present the main screens of the implemented mobile application.

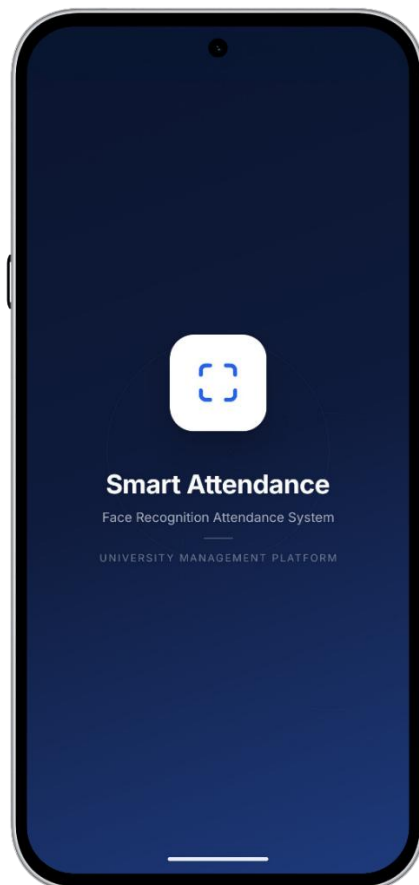


Figure 5.1 Splash Screen

This screen appears when the application starts and displays the system identity before loading the main interface.

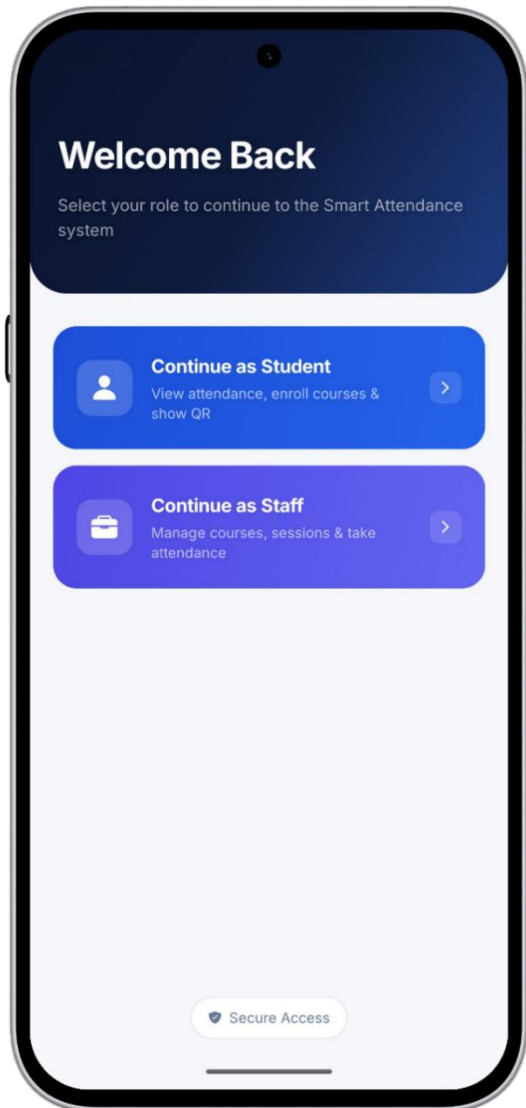


Figure 5. 2 Role Selection Screen

This screen allows users to select their role (Student or Academic Staff) before accessing the system.

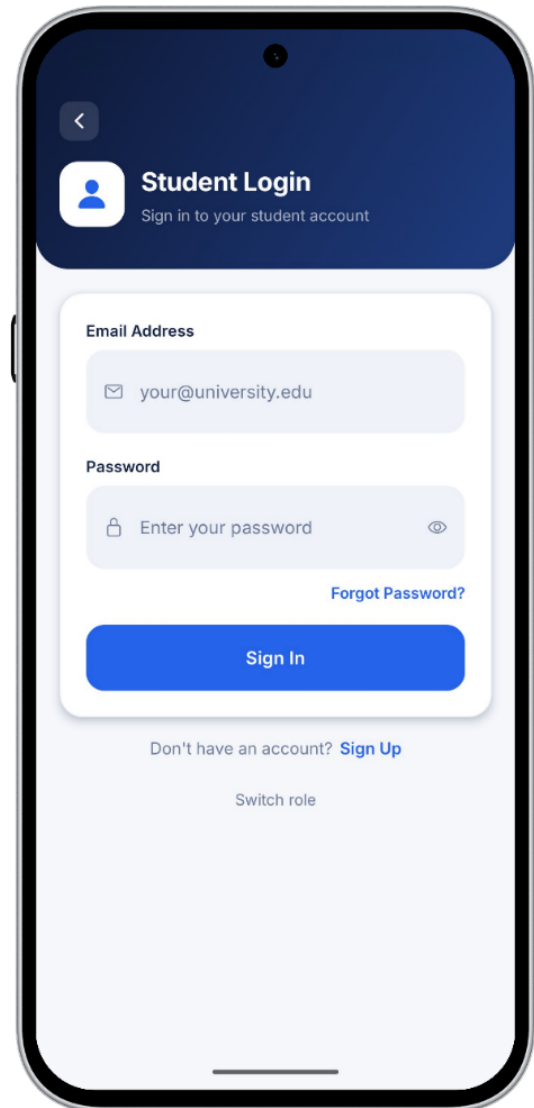


Figure 5. 3 Student Login Screen

Students use this screen to authenticate and access their personal dashboard.

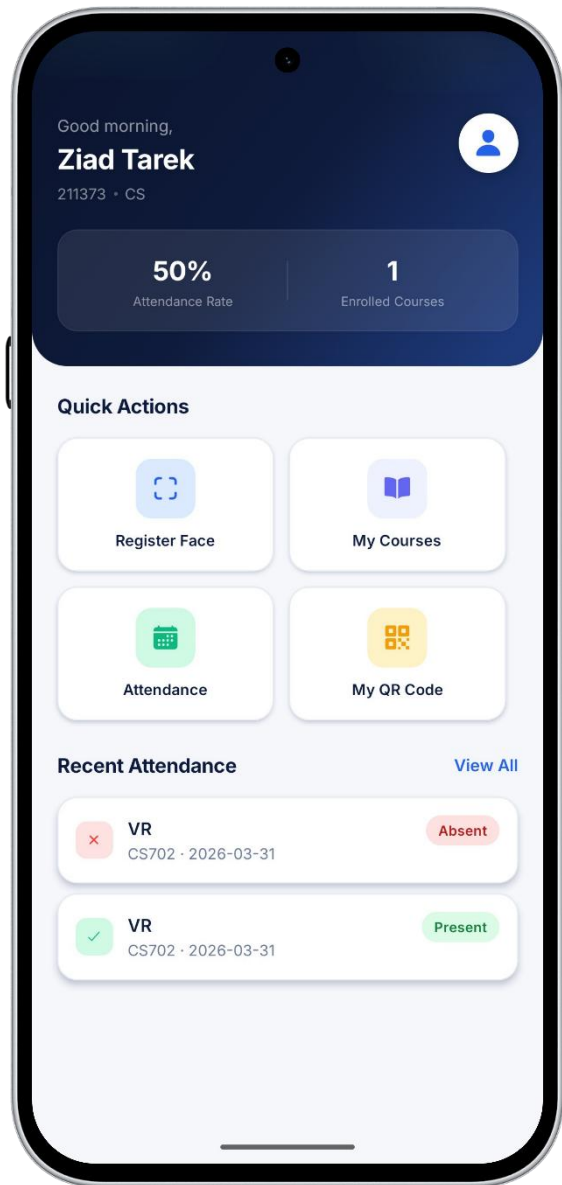


Figure 5. 4 Student Dashboard

The dashboard provides students with attendance statistics, enrolled courses, quick access to system functions, and recent attendance records.

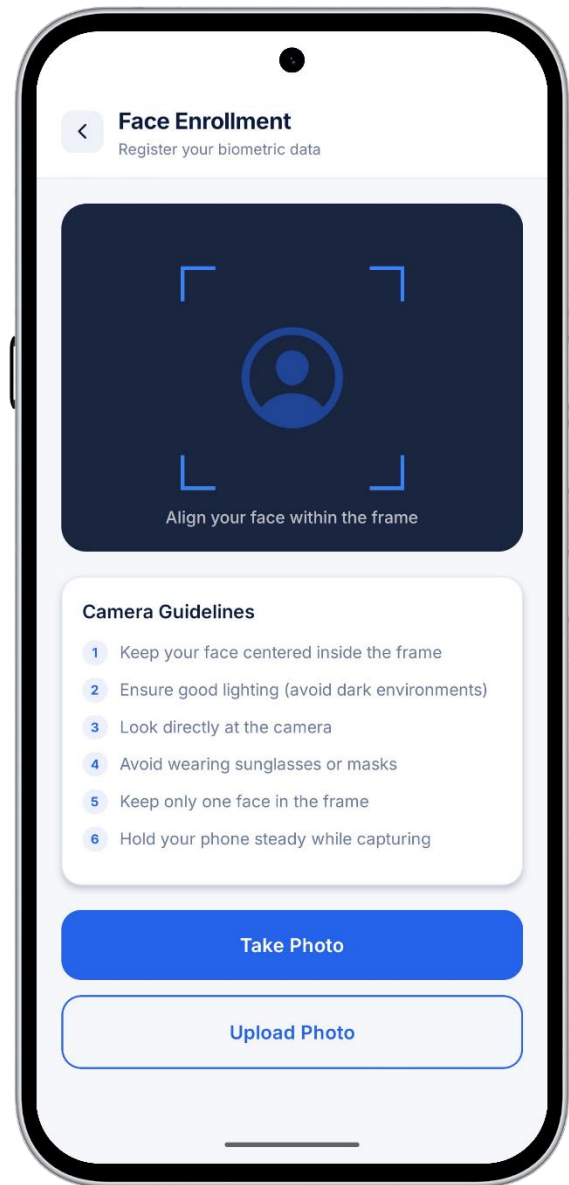


Figure 5. 5 Face Enrollment Screen

Students register their facial data through this screen, which is later used for attendance verification.

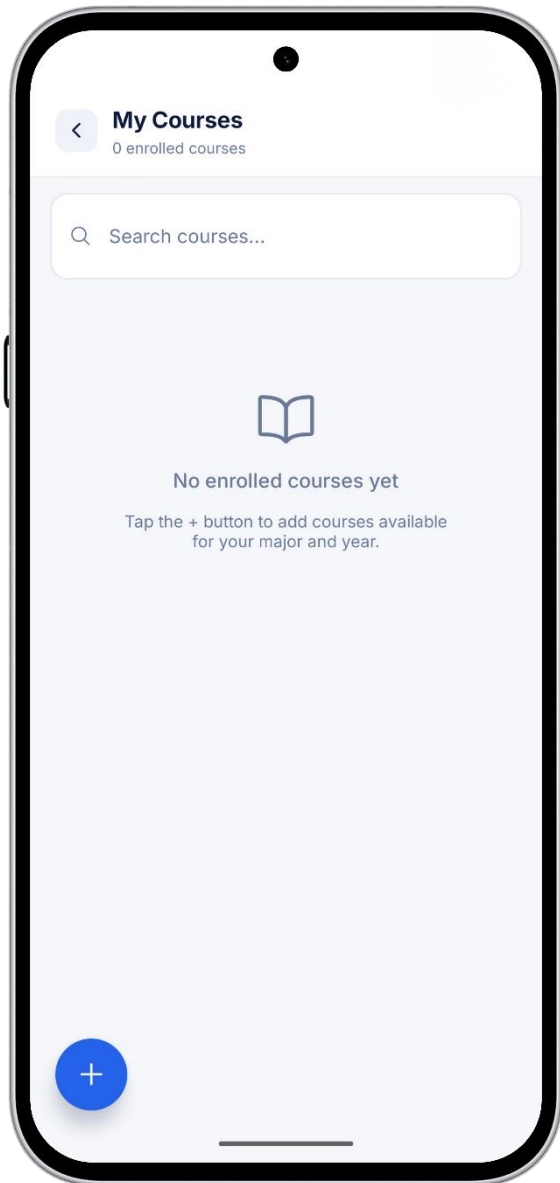


Figure 5. 6 Course Enrollment Screen

Students can browse and enroll in available courses through this interface.

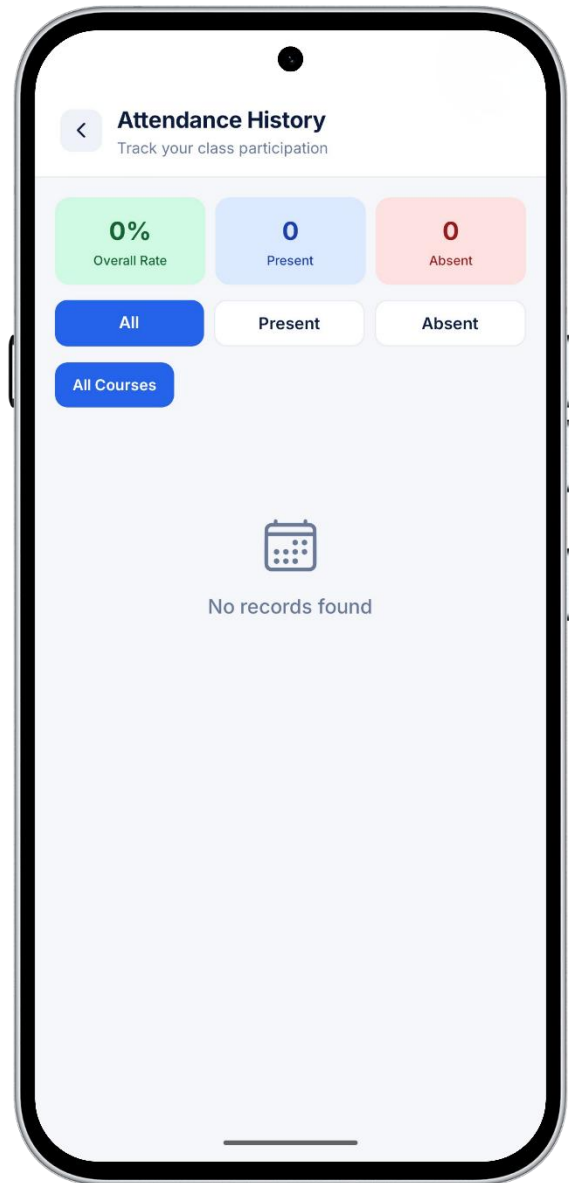


Figure 5. 7 Attendance History Screen

Students can view attendance statistics and track their attendance records for all registered courses.

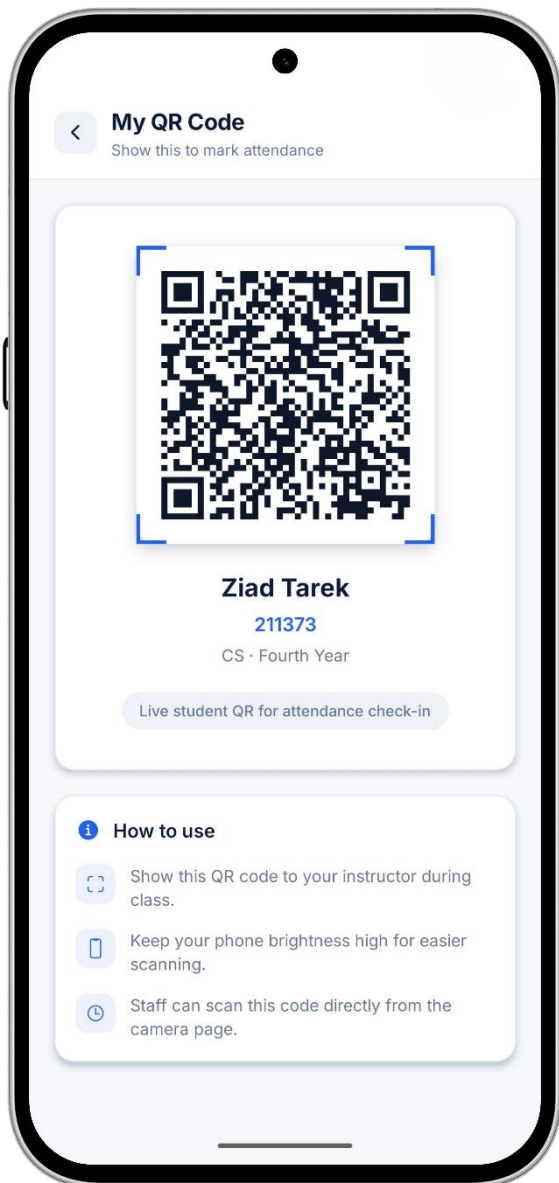


Figure 5. 8 Student QR Code Screen

This screen displays the student's unique QR code used for attendance verification.

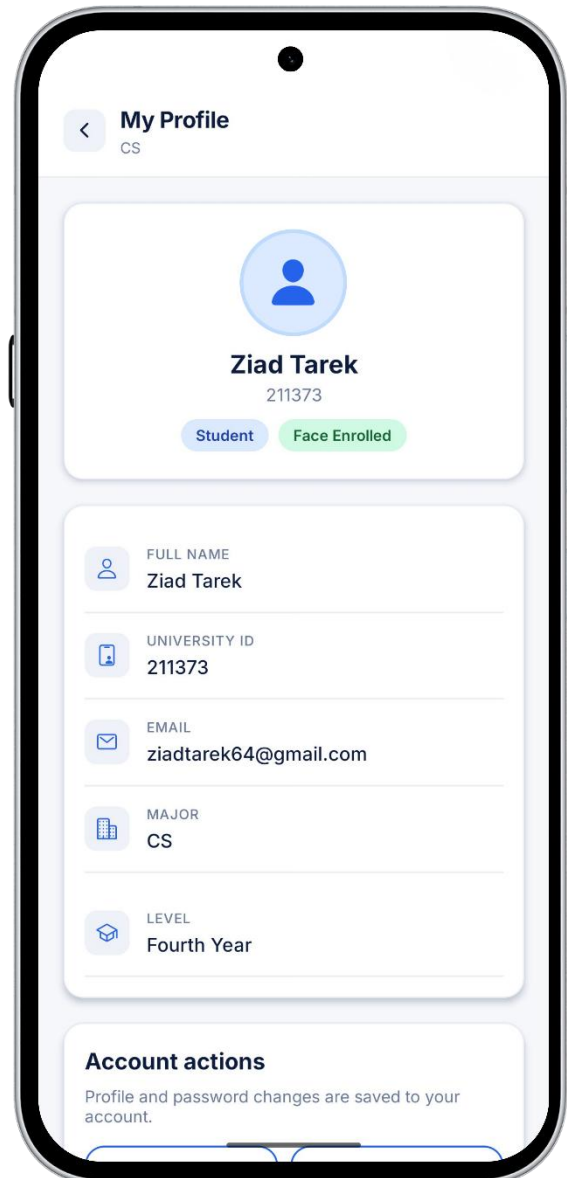


Figure 5. 9 Student Profile Screen

Students can view and manage their personal information, account settings, and facial enrollment status.

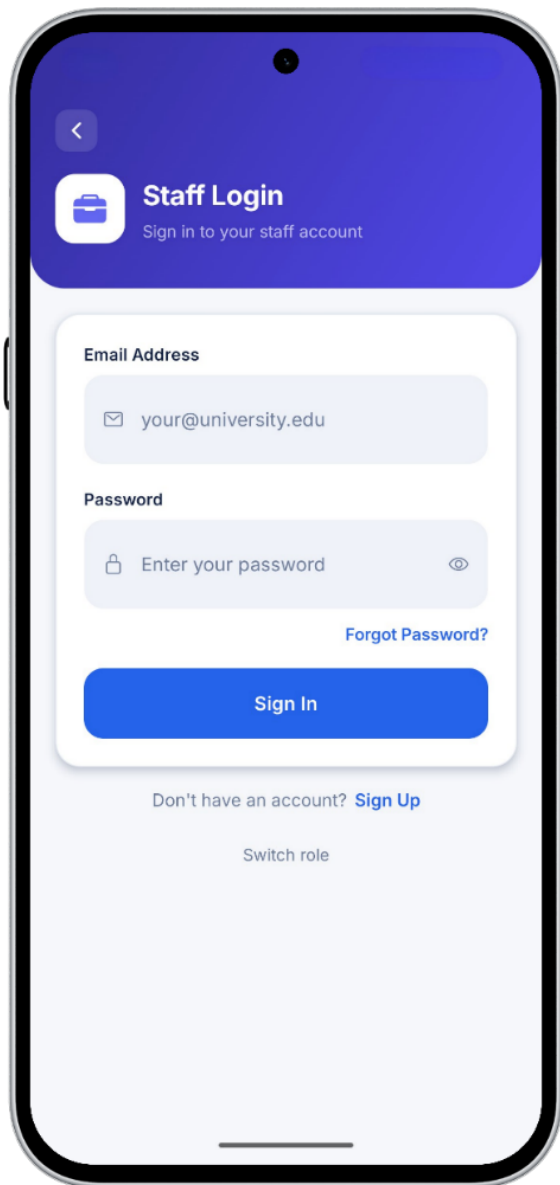


Figure 5. 10 Academic Staff Login Screen

Academic staff authenticate through this screen to manage courses, sessions, and attendance records.

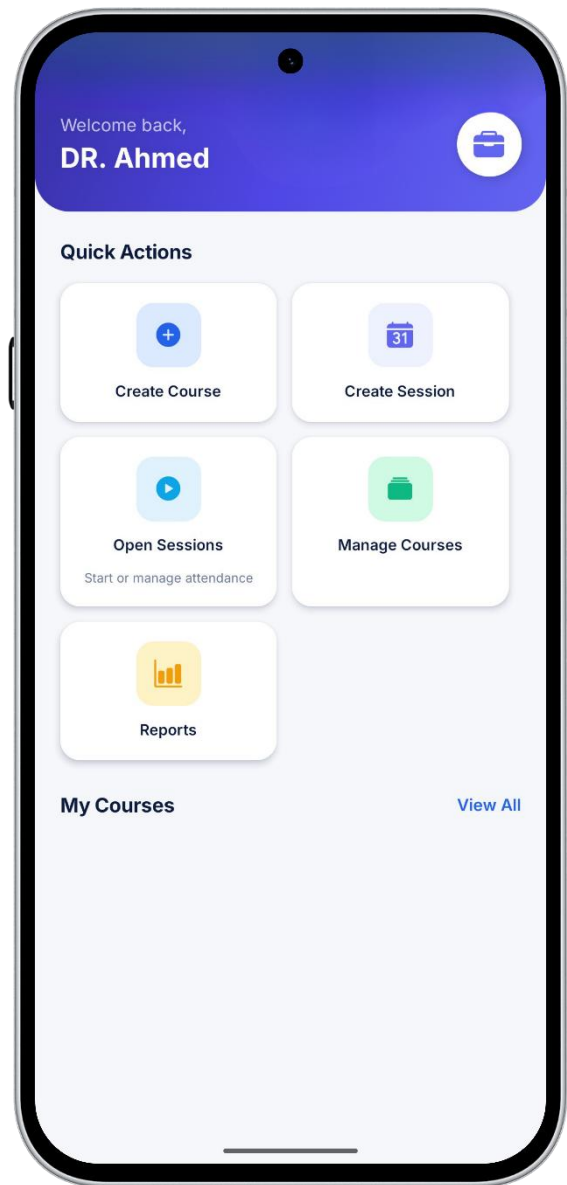


Figure 5. 11 Academic Staff Dashboard

The Academic Staff Dashboard provides quick access to the main system functions, including course management, attendance session management, and report generation

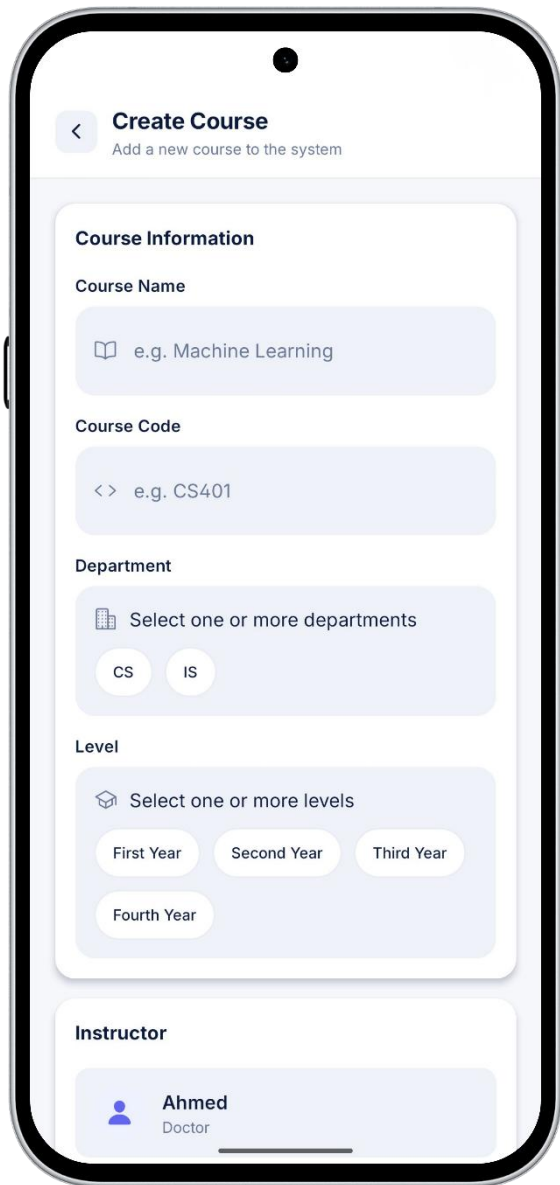


Figure 5.12 Staff Course Creation Screen

This screen allows academic staff to create a new course by entering course information such as the course name, code, department, and academic level.

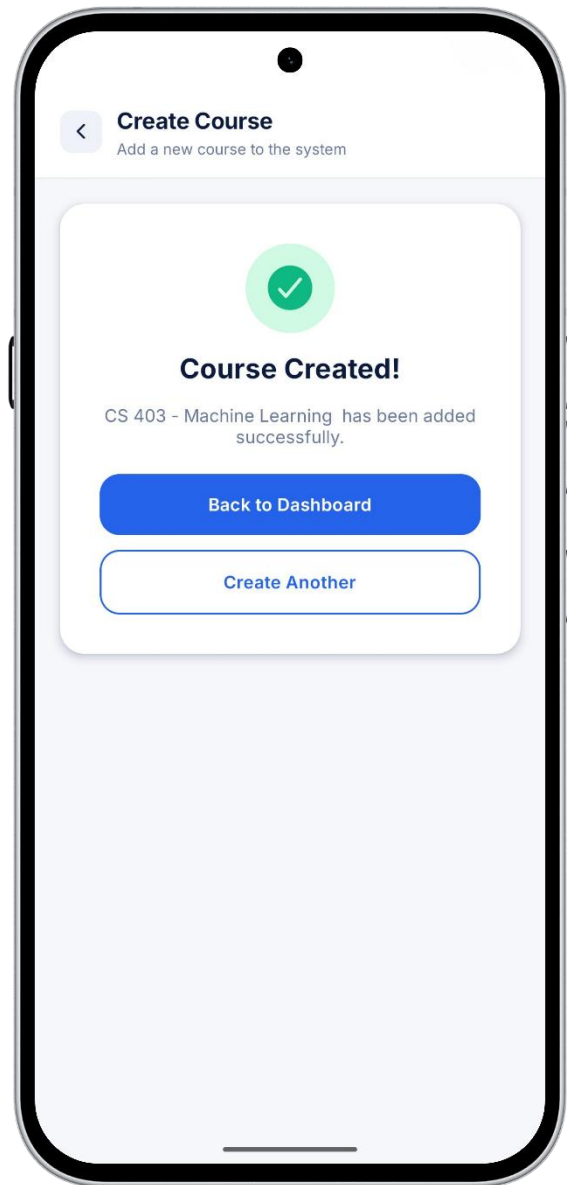


Figure 5.13 Staff Course Creation Screen

This screen confirms the successful creation of a new course and provides options to return to the dashboard or create another course.

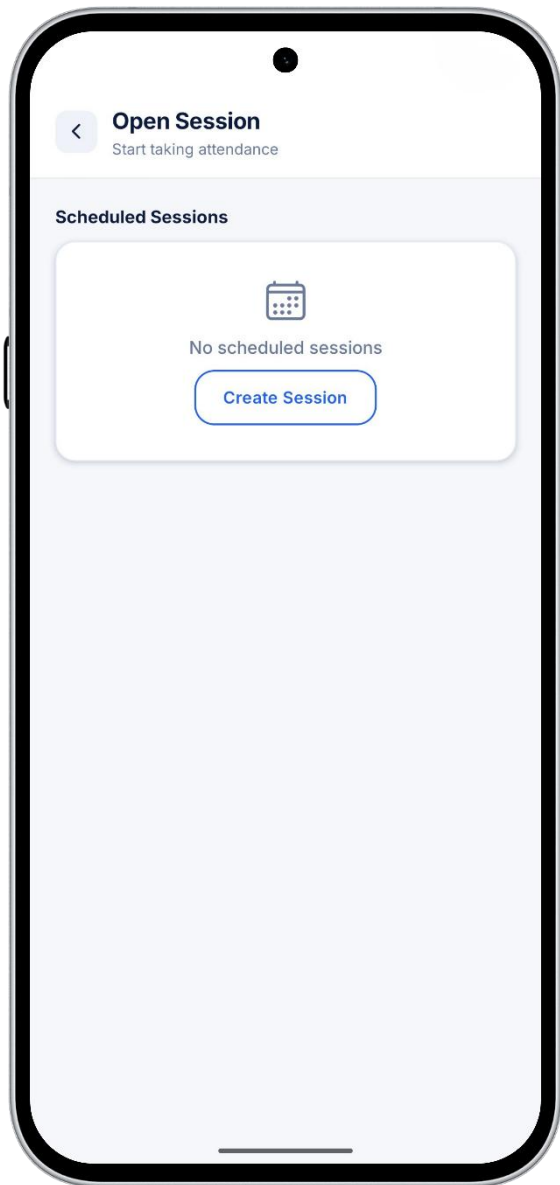


Figure 5. 14 Staff Open Session Screen

This screen enables academic staff to view available attendance sessions and create a new session when required.

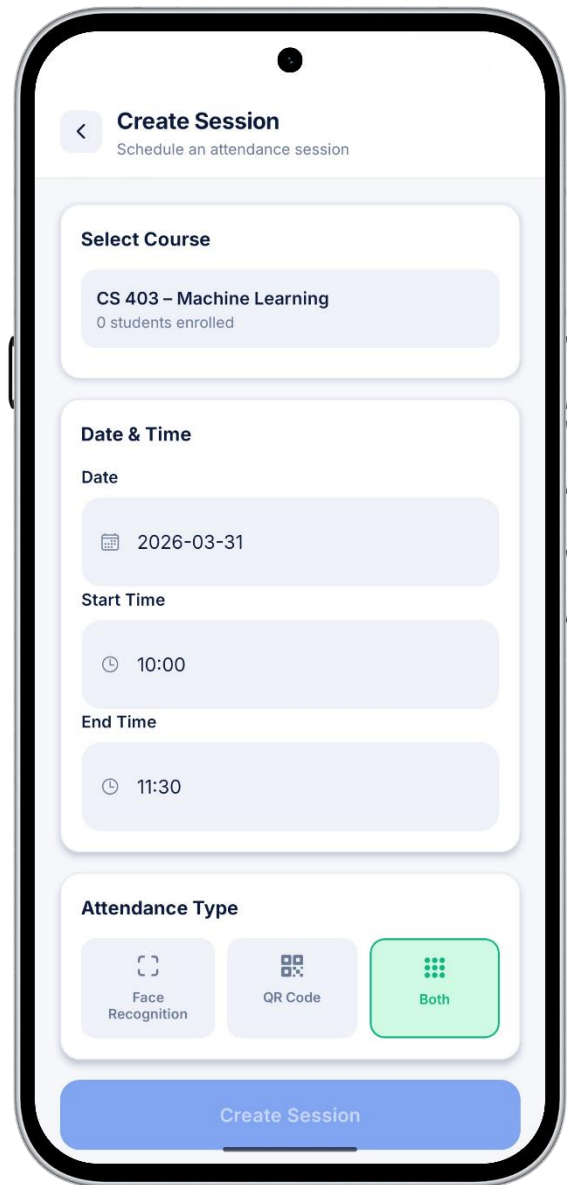


Figure 5. 15 Staff Session Creation

This screen allows academic staff to create an attendance session by selecting a course, specifying the date and time, and choosing the attendance method.

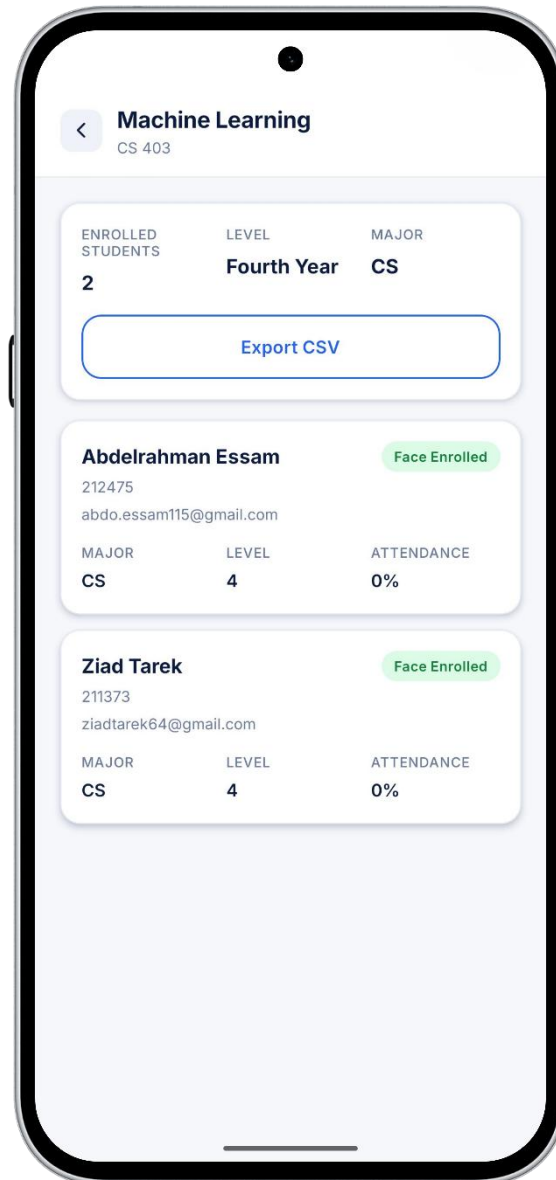


Figure 5. 16 Staff Export Attendance

This screen displays enrolled students and attendance information for a selected course, while providing the ability to export attendance records for reporting purposes.

5.3 Results and Discussion

The implementation of the proposed **Smart Face Recognition Attendance System** successfully achieved the objectives defined during the analysis and design phases. The developed system provides an efficient and secure solution for managing student attendance by replacing traditional manual methods with a digital and automated approach.

The integration of **Face Recognition** and **QR Code** technologies enabled accurate and flexible attendance recording. Students can register their facial data and generate a unique QR code, while doctors can verify attendance using either method during an active attendance session. The system also prevents duplicate attendance records and ensures that only enrolled students can be marked as present.

The developed application provides separate functionalities for students and doctors through a role-based access mechanism. Students can register, enroll in courses, view their enrolled courses, and monitor their attendance reports. Doctors can create courses, manage attendance sessions, verify attendance, and generate attendance reports.

The backend APIs, developed using **Laravel**, communicated successfully with the **React Native** mobile application, while the **MySQL** database efficiently managed users, courses, enrollments, attendance sessions, and attendance records. The modular architecture improved system maintainability and supports future enhancements.

Overall, the obtained results demonstrate that the proposed system is reliable, user-friendly, and capable of simplifying the attendance management process while reducing administrative effort and minimizing human errors. The implementation confirms the effectiveness of combining modern mobile technologies with intelligent attendance verification methods to support digital transformation in educational institutions.

Chapter 6: Testing

6.1 Unit Testing

Unit Testing was performed to verify that each individual component of the **Smart Face Recognition Attendance System** functions correctly and independently before integrating it with the rest of the system. The purpose of this testing phase was to detect errors at an early stage and ensure that every module produces the expected output for valid and invalid inputs.

The testing process focused primarily on the backend APIs developed using **Laravel**. Each endpoint was tested individually using **Postman** to validate request handling, input validation, authentication, authorization, and response accuracy. The tested modules included user registration, login, password recovery, course management, student enrollment, attendance session management, face registration, face verification, QR code generation and scanning, and attendance reporting.

In addition to API testing, the complete mobile application was evaluated through **user testing**. After the implementation phase was completed, **six testers** were asked to use the application and perform its main functionalities, including registration, login, course enrollment, attendance recording, QR code scanning, face recognition, and report viewing. Their feedback was used to identify minor usability issues and verify that the system behaved as expected under normal usage conditions.

Furthermore, validation rules and error-handling mechanisms were tested to ensure that invalid requests, duplicate operations, and unauthorized access attempts were handled correctly. The results confirmed that each module operated as expected and returned appropriate status codes and responses.

The successful completion of both **technical testing using Postman** and **practical user testing by six testers** increased the reliability of the system and provided a solid foundation for the integration of all system components during the next testing phase.

Testing Method	Description	Result
API Testing	Tested all REST APIs using Postman	Passed
Functional Testing	Tested system functionalities	Passed
User Testing	Six testers evaluated the application	Passed
Validation Testing	Checked invalid and duplicate inputs	Passed
Authentication Testing	Verified role-based access control	Passed

Table 6.1 Unit Testing

6.2 Integration Testing

Integration Testing was conducted after the successful completion of Unit Testing to verify that all system modules work together correctly and exchange data without errors. The objective of this phase was to ensure that the interaction between the mobile application, backend APIs, database, and external services operates as expected.

The testing process involved integrating the **React Native** mobile application with the **Laravel** backend through RESTful APIs while using the **MySQL** database for data storage and retrieval. Different scenarios were executed to verify the complete workflow of the system.

Several integrated operations were tested, including user registration and login, course creation by doctors, student enrollment, face registration, QR code generation, attendance session creation, attendance recording through both **Face Recognition** and **QR Code**, report generation, and attendance percentage calculation. The system was also tested to ensure that only enrolled students could record attendance and that duplicate attendance records were prevented.

Furthermore, the communication between the backend and the external **Face Recognition API** was verified to ensure accurate face matching and proper response handling. The interaction between different modules was monitored to confirm that changes in one component were correctly reflected throughout the entire system.

Finally, the fully integrated application was evaluated by **six testers**, who performed complete end-to-end scenarios using both student and doctor accounts. Their testing confirmed that the modules interacted correctly, the application functioned smoothly, and the overall workflow met the intended system requirements.

The results of Integration Testing demonstrated that all components of the Smart Attendance System operate cohesively, providing a reliable, secure, and efficient solution for attendance management.

Integration Scenario	Components Tested	Result
Register → Login	Mobile App + API + Database	Passed
Create Course → Enroll	API + Database	Passed
Create Session → Face Verification	Mobile App + API + Face Recognition Service	Passed
Create Session → QR Scan	Mobile App + API + QR Module	Passed
Attendance → Report Generation	Database + Reporting Module	Passed
Complete End-to-End Workflow	All System Components	Passed

Table 6.2 results of Integration Testing

Conclusion

The Smart Face Recognition Attendance System presented in this project provides a modern and efficient solution for managing student attendance by replacing traditional manual methods with a secure and automated digital system. By integrating **Face Recognition** and **QR Code** technologies, the system improves the accuracy, speed, and reliability of the attendance process.

The system was designed using a client-server architecture that separates the backend from the mobile application, ensuring flexibility, scalability, and easier maintenance. It includes features such as **user authentication, course management, student enrollment, attendance session management, automatic attendance recording,** and **report generation**, effectively addressing the limitations of conventional attendance systems.

In addition, the reporting module enables students to monitor their attendance percentages while allowing doctors to generate detailed attendance reports. The database design and API architecture ensure data consistency and support future expansion, while role-based access control enhances system security by restricting access according to user responsibilities.

Overall, the proposed **Smart Face Recognition Attendance System** achieves its objectives by providing an **accurate, secure, and user-friendly attendance management solution**. It reduces administrative effort, minimizes human errors, prevents fraudulent attendance, and supports the digital transformation of educational institutions.

Future work

Although the proposed **Smart Face Recognition Attendance System** successfully achieves its main objectives, there are several opportunities for future improvements that can enhance its functionality, security, and user experience.

One possible enhancement is the integration of **real-time notifications**, allowing students to receive instant alerts about attendance confirmation, upcoming lectures, or important course announcements. Similarly, doctors could receive notifications regarding attendance statistics and session updates.

Another valuable improvement is the implementation of **multi-face recognition**, enabling the system to identify and record the attendance of multiple students simultaneously from a single image or live camera feed. This feature would significantly reduce the time required for attendance taking in large classrooms.

The system can also be expanded by integrating **GPS and geofencing technologies** to ensure that attendance is recorded only when students are physically present within the university or classroom boundaries. This would provide an additional layer of security against fraudulent attendance.

Furthermore, the reporting module can be enhanced by generating **PDF reports**, advanced statistical dashboards, and visual analytics that help doctors and administrators evaluate student attendance trends throughout the academic semester.

Another future enhancement is the development of an **administrative dashboard** that enables university management to monitor courses, departments, students, and attendance statistics from a centralized platform with different access privileges.

The project can also benefit from integration with existing **University Management Systems (UMS)** or **Learning Management Systems (LMS)** to automatically synchronize student information, course registrations, and academic schedules.

Finally, deploying the system on a **cloud-based infrastructure** with support for scalable databases and secure backup mechanisms would improve system availability, reliability, and performance while making it accessible to multiple faculties and universities.

These future enhancements would transform the proposed **Smart Face Recognition Attendance System** into a comprehensive smart campus solution capable of supporting the continuous digital transformation of educational institutions.

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Appendices

The appendices contain supplementary materials that support the development, implementation, and evaluation of the proposed Smart Face Recognition Attendance System. These materials provide additional technical details, screenshots, diagrams, and testing evidence that complement the information presented in the main chapters of this report.

Appendix A: User Interface Screenshots

This appendix contains high-resolution screenshots of the mobile application interfaces for both students and academic staff, including authentication screens, course management, attendance management, QR code generation, face registration, and reporting screens.

Appendix B: Database Design

This appendix presents the Entity Relationship Diagram (ERD), database schema, and table relationships used in the implementation of the system.

Appendix C: API Documentation

This appendix includes the REST API endpoints developed for the system, along with request methods, parameters, authentication requirements, and response formats.

Appendix D: Testing Results

This appendix contains testing evidence, sample test cases, Postman API testing results, and screenshots demonstrating the successful operation of the system.

Appendix E: Project Repository

The complete source code, user interface resources, and supporting project materials are available in the project's GitHub repository:

GitHub Repository:

<https://github.com/Diiezel/Smart-Face-Recognition-Attendance-System>

ملخص المشروع بالعربي

يهدف هذا المشروع إلى تصميم وتطوير نظام ذكي لإدارة حضور الطلاب (Smart Attendance System) يعتمد على التقنيات الحديثة بدلاً من الطرق التقليدية لتسجيل الحضور، وذلك من خلال استخدام تقنية التعرف على الوجه (Face Recognition) ورموز الاستجابة السريعة (QR Code)، مما يساهم في زيادة دقة وسرعة عملية تسجيل الحضور وتقليل الأخطاء البشرية ومحاولات التلاعب.

يتكون النظام من تطبيق مخصص للطلاب وأعضاء هيئة التدريس، بالإضافة إلى خادم (Backend) مسؤول عن إدارة البيانات والعمليات المختلفة. يتيح النظام للطلاب إنشاء حسابات شخصية، وتسجيل المواد الدراسية، وتسجيل بصمة الوجه الخاصة بهم، والاطلاع على نسبة حضورهم وتقاريرهم الأكاديمية. كما يتيح لأعضاء هيئة التدريس إنشاء المقررات الدراسية، وإدارة جلسات الحضور، وتسجيل حضور الطلاب باستخدام تقنية التعرف على الوجه أو من خلال مسح رمز الـ QR الخاص بكل طالب، بالإضافة إلى استخراج تقارير الحضور بصيغة Excel.

تم تطوير النظام باستخدام **Laravel** لبناء واجهات برمجة التطبيقات (RESTful APIs)، مع تصميم قاعدة بيانات علائقية لإدارة المستخدمين والمقررات وجلسات الحضور وسجلات الحضور. كما تم تطبيق نظام صلاحيات (Role-Based Access Control) لضمان أن يتمكن كل مستخدم من الوصول إلى الوظائف المخصصة له فقط، سواء كان طالباً أو عضو هيئة تدريس.

يساهم النظام في تقليل الوقت والجهد اللازمين لتسجيل الحضور، وتحسين دقة البيانات، وتوفير تقارير إحصائية تساعد أعضاء هيئة التدريس والطلاب على متابعة نسب الحضور بشكل مستمر. كما أن تصميم النظام يسمح بإضافة مزايا مستقبلية مثل التعرف على عدة وجوه في وقت واحد، وإصدار تقارير بصيغة PDF، وربط النظام بأنظمة إدارة الجامعات المختلفة.

وبذلك يقدم المشروع حلاً عملياً وحديثاً لإدارة حضور الطلاب، ويواكب الاتجاه نحو التحول الرقمي واستخدام تقنيات الذكاء الاصطناعي في المؤسسات التعليمية، بما يساهم في رفع كفاءة العملية التعليمية وتحسين جودة الخدمات المقدمة.